

Applied Research Methods

Course code: MGT-808

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Research Concepts/Meaning of Research

- Meaning: Research is an art of scientific investigation which refers to a search of knowledge. Research is also systematized effort to gain new knowledge. Research is an academic activity and is actually a journey of discovery.
- So, Research is an original contribution to the existing stock of knowledge (in various fields/spheres of activities) for its advancement.
- The term Research also refers to the systematic method (s) consisting of identifying the problem, formulating hypothesis, collecting the facts or data, analysing the data & reaching to certain conclusion(s).

- **Significance of research:** Following are the significance of conducting a research:
- (i) Research provides the basis for nearly all govt. policies in our economic/Business/trade system. Through research, one can advise alternative policies and can as well examine the consequences of each of these alternatives.
- Research has its special significance in solving various operational & planning problems of business & industry. Operational research refers to the application of mathematical, logical & analytical techniques to the solution of business problems of cost minimization or of profit maximization or what can be termed as optimization problems.

- Research is equally important for social scientists in studying social relationship and in seeking answers to various special problems. In addition to these, the significance of research can also be understood keeping in view the following points
- To those students who are to write a Master's or Ph.D thesis, research may mean a careerism or a way to attain high position in the social structure
- To professionals, research may mean a source of livelihood.
- To analyst and intellectuals, research may mean the generalization of new theories.

- Thus, research is the foundation of knowledge for the sake of knowledge and important source of providing guidelines for solving different business, governmental and social problems. It is a sort of formal training which enables to understand the new development in one's field in a better way.
- MOTIVATION IN RESEARCH:
- It is often asked that why the people conduct Research? This question is of fundamental importance and some of the possible motives of research may be one or more of the following:
- (i) Desire to get a research degree/certificate along with its consequential benefits.

- Desire to get intellectual joy.
 - Desire to be of service to society.
 - Desire to get respectability.
 - However, this is not a complete list of factors motivating people to undertake research studies.
 - **Major Types of Research**
 - (i) Applied vs Fundamental research.
 - (ii) Quantitative vs Qualitative research.
 - (iii) Business vs Social research.
- These are defined as below:
- (i) Research can either be applied (or action) research or fundamental (basic or pure) research.

- Applied research aims at solving an immediate (current) problem facing a society or a business/ industrial organisation. Whereas fundamental research is concerned with generalization and formulation of a theory.
- In simple words, gathering knowledge for knowledge's sake is called 'pure' or 'basic' research; while research concerning some basic phenomenon or relating to pure Mathematics are examples of fundamental research.
- Similarly, research studies concerning human behavior carried out with a view to make generalizations about human behavior are also

- examples of fundamental research, but research aimed at certain conclusions (say, a solution) facing a concrete social or business problem is an example of applied research.
- Both basic & applied research have to be carried out in a scientific manner so that the findings or results generated by them can be relied upon to effectively solve the problem investigated. It is, however, possible that some applied research may have some shorter time frame than some basic research.

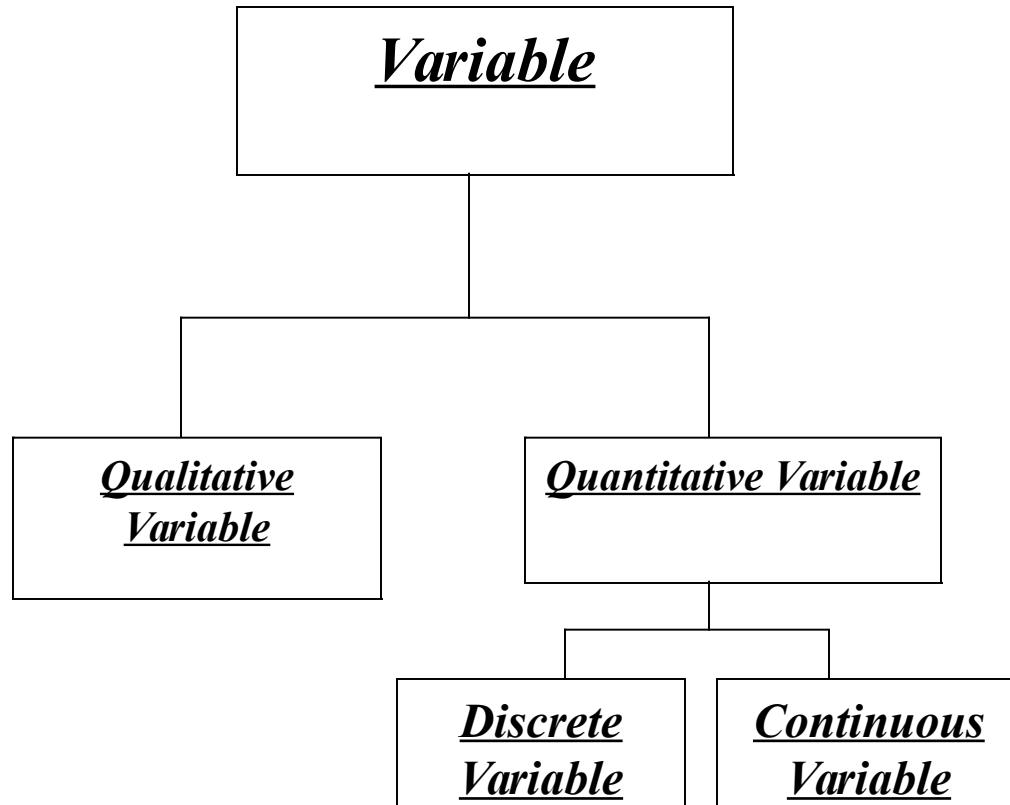
- (ii) Quantitative vs Qualitative Research: Quantitative research is based on measurement of quantity or amount. It is applied to phenomena that can be expressed in terms of quantity. Quantitative research attempts precise measurement of something.
- In business research, quantitative methodologies usually measure consumer behavior, knowledge, opinions, or attitudes. Such methodologies answer questions related to how much, how many, how often, when and who. Note survey is considered a dominant methodology of quantitative research.

- On the other hand, Qualitative research is concerned with qualitative phenomenon, i.e. a phenomenon relating or involving quality or kind. Qualitative research includes, perception of people towards like & dislike of any brand, word association tests and similar other projective techniques.
- (iii) Business and Social Research:
Business research can be described as a systematic and organised effort to investigate a specific problem encountered in the work setting, which needs a solution. It comprises a series of steps that are designed & executed with the goal of finding

- Answers to the issues that are of concern to the Manager in the work environment. Specifically, business research can be defined as an organised, systematic, data-based, objective scientific inquiry, or investigation into a specific problem, undertaken with the purpose of finding answers or solutions to it.
- In essence, research provides the necessary information that guide Managers to make informed decisions to successfully deal with problems. The information provided could be the result of a careful analysis of data gathered first-hand or of data that are already available (it may be noted that data can be quantitative or qualitative

- (in the company, industry, archives, etc.). It may be noted that data can be quantitative or qualitative.
- **Generally business research can be undertaken for two purposes:**
 - 1. To solve a current problem faced by the Manager in the work setting that is called applied research.
 - 2. To generate a body of knowledge about how to solve problems that could be occurred in organisations. This is called basic research or fundamental research. It is also known as pure research.
- Example of hypothesis: Experience is positively related to job performance.

Definition of Various Concepts used in Research



Definition of Various Concepts used in Research

Variable:

Any characteristic of interest which takes on different values is called variable. For example: production, cost, temperature, sale of a market, consumption, price of a commodity at different locations etc. Variable is broadly divided into qualitative and quantitative variables.

Qualitative and Quantitative Variables

A variable is defined to be qualitative which is not capable of numerical measurement but one can feel the presence or absence of a particular phenomena. For example, honesty, beauty, race, like and dislike, pass or fail, gender classification etc.

A variable is defined to be quantitative which is capable of numerical measurement. For example, cost of production, price of a commodity, monthly consumption of households etc.

Definition of Various Concepts used in Research

Discrete and Continuous Variables

A variable is said to be discrete if it takes isolated integral values or a variable which take the values on jumps is called a discrete variable. For example, number of rooms in a house, number of students in the class, number of Banks in different cities, size of a household, number of shops in a market etc.

A type of variable which takes all possible values with in a given interval/range (a, b) . For example, consumption, production, temperature, monthly sale of a market, height, weight and age etc.

Classify the variables as discrete or continuous

- (i) The amount of milk produced by a cow.
- (ii) The number of heads in the toss of 6 coins.
- (iii) The life times of television tubes produced by Sony.
- (iv) The number of petals on a flower.
- (v) Hourly temperature of Islamabad recorded by Met. Office.
- (vi) Total eggs laid by a hen in a month.
- (vii) Amount of CNG/petrol filled in cars at a pump.
- (viii) The yearly income of a university professor.
- (ix) Time taken to score the first goal in a football match
- (x) No. of goals scored in each of 50 football matches.

Other Variables in Research

1. Dependent Variable
2. Independent Variable
3. Moderating Variable
4. Mediating or Intervening Variable

Other Variables in Research

Dependent and Independent Variables

A type of variable which is influenced by other variable/variables is called dependent variable. It is also called random or stochastic variable. OR

A variable which depends on one or more other variables is called dependent variable. OR

A variable of primary interest that lends itself for investigation as a function of other cause variables is known as dependent variable.

Other Variables in Research

For example, in economics, consumption of a commodity (say apple) depends upon the income, household size, and price etc of the commodity.

In this example, consumption of apple is a dependent variable which will vary from one family to other family; while the other variables like income, household size and price are independent variables.

A variable which influence a dependent variable in either direction (positive or negative) is called independent variable.

Other Variables in Research

Moderating Variable:

A variable which modify the original relationship is called moderating variable.

Or

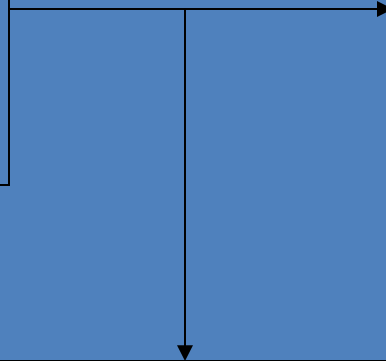
A variable that has a strong contingent or contributory effect on the relationship between dependent and independent variables, is called moderating variable.

For example, Amount of study in terms of study hours affect the grades, however, those student consulting quality information for the purpose may even perform much better as compared to others students. In this example, it is evident that quality information is a moderating variable which is not considered in the research study but have contributory effect on the final grades, so called moderating variable.

Amount of
Study hours
(Independent var.)

Final grade
Test score
(Dependent var.)

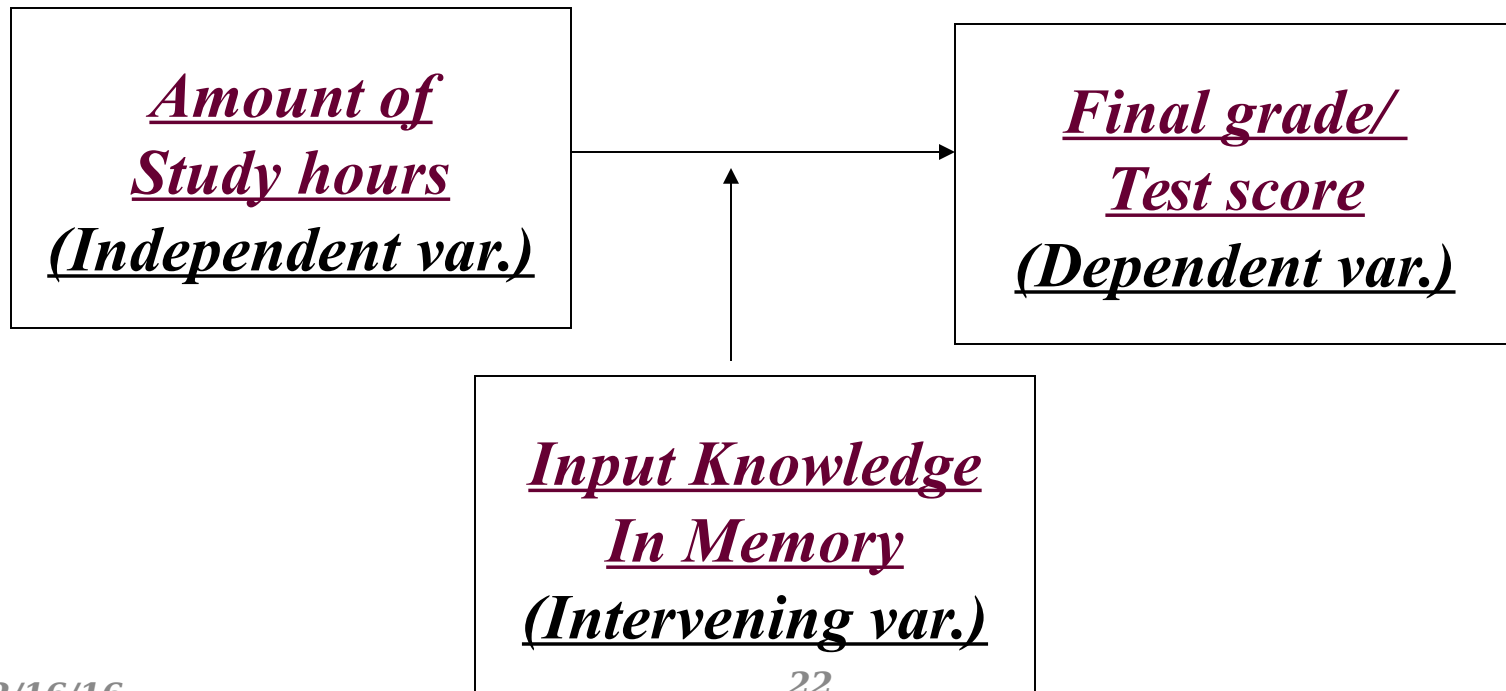
Quality
Information
(Moderating var.)



Other Variables in Research

Intervening Variable:

A variable that surfaces between the time the independent variable starts operating to influence the dependent variable and its impact is felt on it, is called intervening variable.



Research Process

- Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps. It may be noted that the number of steps described by different authors/researchers varies between 6 to 10. However, research process involves a sequence of highly inter-related activities, therefore, research often follows a general pattern. The following steps/stages will describe a research process.
- 1. Define research problem/Research objectives.
- 2. Review the literature or Literature survey.

- 3. Formulate hypotheses.
- 4. Research Design (including sample design).
- 5. Collecting the data (execution).
- 6. Analysing the data (Test hypotheses if any).
- 7. Interpret and report.

- **1. Research Problem:**

- In research process, the first & foremost step happens to be that of selecting & properly defining a research problem. Like a medical doctor, a researcher must examine all the symptoms (presented to him or observed by him) concerning a problem before he can diagnose correctly.

- It is pertinent to mention that scientific research generally starts with a broad area problem area with a definite aim or purpose. In business, the research process involves translating the business decision situation into specific research objectives.
- The broad problem area is identified through the process of observing & focusing on the specific situation. A problem does not necessarily imply that something is seriously wrong with a current situation that needs to be rectified. Some examples of broad area problem are: Frequent production interruptions, lack of interest of employees in their work, training program are perhaps not as effective as anticipated or the sales volume of a product is dropping and so on.

- To find solutions for identified problems, a problem statement that states the general objective of the research should be developed. It means once we have identified the broad area problem area, it needs to be narrowed down to a specific problem statement after some preliminary information is gathered by researcher.
- For example, if it is observed that a company is losing customers, the preliminary information gathering involves the seeking of information in depth, of what is observed. This could be done by a literature review (literature on customer switching) or by talking to several people in the work setting, to clients (why do they switch), or to other relevant sources, thereby gathering information on what is happening and why.

- Thus we get an idea of the situation and consequently this allows us to develop a specific problem statement.
- The research problem undertaken for study must be carefully selected. Help may be taken from a research guide in this connection. Nevertheless, every researcher must find his own salvation for research problem cannot be borrowed. A research problem must spring from the researcher's mind like a plant springing from its own seed. If our eyes need glasses, it is not the optician alone who decides about the number of lens we require. We have to see ourselves & enable him to prescribe for us the right number by cooperating with him. Thus, a research guide can at the most only help a researcher choose a subject.

- It is desirable to make a good problem statement because problem statement introduces the key problem that is addressed in the research project. A problem statement is clear, precise & to the point statement of the specific issue that a researcher wishes to investigate. There are three key criteria to assess the quality of the problem statement: it should be relevant, feasible & interesting. A few examples of well-defined problem statement are:
- Has the social media advertising message resulted in enhanced recall?
- What are the effects of downsizing on the long-range growth patterns of companies?
- How has the new packaging affected the sales of the product?

Literature Review

- Once the problem is formulated, a brief summary of it should be written down. As a general rule, researchers should first investigate previous research to see whether or not other may have already addressed similar research problems. This task can be carried out by literature review, which is a step by step process that involves the identification of published & unpublished work from secondary data sources on the topic of interest. A literature review should help the researcher to identify & highlight the important variables that are related to the problem.
- Note that a literature review helps the researcher to develop a good problem statement: it ensures that no

- Important variable is overlooked in the process of defining the problem. A review of the literature also serves some other functions. For example, sometimes the researcher might spend considerable time & effort in discovering something that has already been thoroughly researched. A lit. review would prevent such a waste of resources. It also gives a good basic frame work to proceed further with the investigation.
- A good lit. review thus provides the foundation for developing a comprehensive theoretical framework from which hypotheses can be developed for testing.
- **Conducting the Lit. Review:** The first step of the lit. review involves the identification of the various published & unpublished materials that are available on the topic of interest, and gaining access to these.

- Data sources: The quality of lit. review depends on a cautious selection & reading of books, academic & professional journals, reports, theses, conference proceedings, unpublished manuscripts and the like. Academic books & journals are, in general, the most useful sources of information. However, other sources such as professional journals, reports, and even news papers may also be valuable because they can provide you with specific, real-world information about markets, industries, or companies. Therefore, as a rule, you will need to use a combination of information resources.
- Text Books: Text books are a useful source of theory in a specific area. An advantage of text books is that they can cover a wide range of topics (less up-to-date than..).

- Journals: Both academic & professional journals are important sources of up-to-date information. Articles in academic journals have generally been peer-reviewed: this means that the articles have been subject to the scrutiny of experts in the same field before being accepted for publication (review articles- very useful).
- Theses: Ph.D theses often contain an exhaustive review of the lit. in a specific area. Most theses include several empirical chapters.
- Conference proceedings: Conference proceedings can be useful in providing the latest research, or research that has not (yet) been published. These are very up-to-date , & for this reason this information source is quite valuable if one is working in a relatively new area.

- Unpublished manuscripts: The APA defines an unpublished manuscript as any information source that is not officially released by an individual, publishing house, or other company. Examples of this may include papers accepted for publication but still “in press”, data from an unpublished study, letters and personal communications (often very up-to-date).
- Reports: Govt. departments & corporations carry out a large amount of research. Their published findings provide a useful source of specific market, industry or company information.
- Newspaper: Newspaper provide up-to-date business information. They are a useful source of specific market, industry, or company information.

- Internet: The amount of information that can be found on world wide web is enormous. You can search for books, journals & journal articles, and conference proceedings as well as for specialized data such as company publications & reports.
- Note that the internet is unregulated & unmonitored.
- Reliability?
- Search engines : Google search, Yahoo can help you to find relevant information.

- Variables in research.
- Examples: (i) Production unit- One worker in the manufacturing department may produce one bolt per minute, a second might produce two per minute,
- (ii) Absenteeism: On Monday two members in the sale dept. may be absent, on Tuesday, five members may not show up for work; on Wednesday, there may be no one absent. Range from '0' to 'all' being ..
- (iii) Motivation: The levels of motivation of members in the class or in a work team might take on varying values ranging from “very low” to “very high”. Thus motivation is a variable.
- Exercises: A researcher wants to increase the performance of bank employees in a particular

- Branch. What is the dep. Var. in this case?

- **Theoretical Framework**

- After conducting the interviews, defining the problem and completing a lit. review, one is ready to develop a theoretical framework. A theoretical fwork is the foundation of research as it is the basis of hypotheses one will develop. A theoretical fwork represents your beliefs on how certain phenomena (or variable) are related to each other (a model) & an explanation of why you believe that these variables are associated with each other (a theory). The process of building a theoretical fwork includes:
 - (i) Introducing definition of the concepts or variables in your model.
 - (ii) Developing a conceptual model that provides a

- descriptive representation of your theory.
- Coming up with a theory that provides an explanation for relationship b/t the variables in your model.
- From the theoret. Fwork, the testable hypotheses can be developed to examine whether your theory is valid or not. The hypothesised relationship can therefore be tested through appropriate statistical analyses. Hence the entire research rests on the basis of theoret. Fwork, i.e. it is the foundation on which the entire research project is based. It is a logically developed, described, and elaborated network of associations among the variables deemed relevant to the problem situation & identified

- through such processes as interviews, observations, p is to elaborate the becomes evident that to arrive at good solution to the problem, one should first correctly identify the problem & then the variables that contribute to it. After identifying the appropriate variables, the next step is to elaborate the network of associations among the variables, so that the relevant hypotheses can be developed and subsequently tested. Based on the results of hypothesis testing, the extent to which the problem can be solved becomes evident. The theoret. fwork is thus an important step in the research process.
- The relationship b/w lit. review & theoret. fwork is that the former provides a solid foundation for developing the latter. That is the lit. review identifies

- the variables that might be important, as determined by previous research findings. The theoret framework represents & elaborates relationship among the variables, explains the theory underlying these relations, and describes the nature & direction of the relationship.
- A good theoret framework identifies & defines the important variables in the situation that are relevant to the problem & subsequently describes & explains the interconnection among the variables. The relationships among the ind. Vars., the dep. Var(s), & if applicable, the moderating & intervening variables are elaborated. Should there be any moderating var(s), it is important to explain how & what specific relationships they moderate. Similarly for mediating..

- It may be noted that a good theoretical framework is not necessarily a complex framework.
- There are three basic features that should be incorporated in any theoretical framework.
- (i) The variables considered relevant to the study should be clearly defined
- (ii) A conceptual model that describes relationship b/w the variables in the model should be given.
- (iii) There should be a clear explanation of why we expect these relationships to exist.

- Hypothesis Development:
- After extensive lit.survey, researcher should indicate in clear terms the working hypothesis(es). Working hypothesis is a tentative assumption made in order to draw out & test its logical or empirical consequences. As such the manner in which research hypotheses are developed is particularly important since they provide the focal point for research. The development of working hypothesis plays an important role.
- Hypothesis should be very specific & limited to piece of research in hand because it has to be tested.

Directional and nondirectional hypotheses

- While describing the relationship between two variables or comparing two groups, terms such as positive, negative, more than, less than, & the like are used, then these are directional hypotheses b/c the direction of the relationship b/w the variables (+ positive/-negative) is indicated. For example, see the following statement:

“The greater the stress experienced in the job, the lower the job satisfaction of employees”.
- Similarly, the nature of the difference b/w two groups on a variable (more than/less than) is postulated as:

“Women are more motivated than men”.

- On the other hand, nondirectional hypotheses are those that do postulate a relationship or difference, but offer no indication of direction of these relationships or differences. In other words, though it may be speculated that there is a significant relationship b/w two variables, we may not be able to say whether the relationship is positive or negative. For example, see the following statement/hypothesis: “There is a relationship b/w job and satisfaction”
- Likewise, even if we can assume that there will be differences b/w two groups on a particular variable, we may not be able to say which group will be more & which less on that variable. For example, see the

- following hypothesis:
“There is a difference b/w the work ethic values of European and Asian employees”.
- Nondirectional hypotheses are formulated either b/c the relationships or differences have never been explored & hence there is no basis for indicating the direction, or b/c there have been conflicting findings in previous research studies on the variable. In some studies a positive relationship might have been found, while in others a negative relationship might have been traced. Hence, the current researcher might only be able to hypothesize that there is a significant relationship, but the direction may not be clear. In such cases, the hypotheses can be stated

- Nondirectionally.
- Whenever the direction of the relationship is known, it is better to develop directional hypotheses.
- **What Is A Hypothesis?** Ordinarily, when one talks about hypothesis, one simply means a mere assumption or some supposition to be proved or disproved. But for a researcher hypothesis is a formal question that he intends to resolve. A hypothesis can be defined as a tentative, yet testable, statement, which predicts what you expect to find in your empirical data. Hypotheses are derived from theory on which your conceptual model is based & are often relational in nature. Several testable statements or hypotheses can be drawn from theoretic. framework.

- **Hypothesis; Null and Alternate Hypotheses :**

It

is pertinent to note that a hypothesis can be defined as a testable statement of the relationship among variables. A hypothesis can also test whether there are differences b/w two groups (or among several groups) with respect to any variable or variables. To examine whether or not the assumed relationship s or differences exist, these hypotheses can be set as propositions as under:

- Employees who are more healthy will take sick leave less frequently.
- If employees are more healthy, then they will take sick leave less frequently.

- Students who receive counselling will show a greater increase in creativity than students not receiving counselling.
- The automobile A is performing as well as automobile B.
- Null and Alternate Hypotheses: If we are to compare method A with B about its superiority & if we proceed on the assumption that both methods are equally good, then this assumption is termed as the null hypothesis. As against this, we may think that the method A is superior or the method B is inferior, it is termed as alternate or alternative hypothesis. A null hypothesis (H_0) is a hypothesis set up to be rejected in order to support an alternate hypothesis,

- labeled H_A . When used, the null hypothesis is presumed true until statistical evidence, in the form of a hypothesis test, indicates otherwise. A null hypothesis should always be precise such as “a drug is ineffective in curing a particular disease” or
- “the difference b/w the two teaching methods is null or zero”
- “advertising does not affect sales”
- “men & women buy equal amounts of shoes”

Moderator & mediator variables

- In general terms, a moderator is a qualitative (e.g. sex, race, class) or quantitative (e.g., level of reward) variable that affects the direction and/or strength of relation b/w an independent & a dependent variable.
- Mediator variable: In general, a given variable may be said to function as a mediator to the extent that it accounts for the relation b/w the predictor & dependent variable. Whereas moderator variables specify when certain effects will hold, mediators speak to how or why such effects occur.
- Another way to think about this issue is that a moderator variable is one that influences the strength of a

- relationship b/w two other variables, and a mediator variable is one that explains the relationship b/w the two other variables.

RESEARCH DESIGN

- After the researcher has formulated the research problem, he or she must develop the research design as part of the research design stage. A research design is a master plan that specifies the methods & procedures for collecting & analysing the needed information.

Meaning of Research Design

Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitutes a research design. **OR**

Research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure. In fact, the research design is the conceptual structure within which research is conducted. It constitutes the blueprint for the collection, measurement and analysis of data. More explicitly, the design decisions happen to be in respect of:

Meaning of Research Design

- (i). what is the study about?
- (ii). why is the study being made?
- (iii). where will the study be carried out?
- (iv). what type of data is required?
- (v). where can the required data be found?
- (vi). what periods of time will the study include?
- (vii). what will be the sample design?
- (viii). what techniques of data collection will be used?
- (ix). how will the data be analyzed?
- (x). in what style will the report be prepared?

Meaning of Research Design

Keeping in view the above (i) to (x), one may split the overall research design into the following parts:

- (a).The sampling design which deals with the method of selecting items to be observed for the given study;
- (b).The observational design which relates to the conditions under which the observations are to be made;
- (c).The statistical design which concerns with the question of how many items are to be observed and how the information and data gathered are to be analyzed; and
- (d).The operational design which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.

Important Features of Research Design

Important features of a research design are:

1. It is a plan that specifies the sources and types of information relevant to the research problem.
2. It is a strategy specifying which approach will be used for gathering and analyzing the data.
3. It also includes the time and cost budgets since most studies are done under these two constraints.

In sum up, a research design must at least contain a clear statement of the research problem, procedures and techniques used for gathering information/data, the population to be studied, and the methods to be used in processing and analyzing the data.

Need for Research Design

Before going to conduct a research, a research design is needed because it facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money.

For example, to construct an economical and attractive construction of a house, one should consult an architect or an expert for making a map so that all the necessary things for its construction are highlighted.

Similarly, we need a research design or a plan in advance of data collection and analysis methods for research project keeping in view the objectives, time and labours required.

Need for Research Design

Therefore, “Research design stands for advance planning of the methods to be adopted for collecting the relevant data and the techniques to be used in their analysis, keeping in view the objective of the research and the availability of staff, time and money”. The research design helps the researcher to organize his ideas in a form whereby it will be possible for him/her to look for flaws and inadequacies. Without proper and accurate research design, it will be difficult to provide a comprehensive review of the proposed study (research).

Features of a Good Design

Following are the features of a good research design:

1. Flexible
2. Appropriate
3. Efficient
4. Economical and so on

Generally, the design which minimizes bias and maximizes the reliability of the collected data and analyzed is considered a good design. The design which gives the smallest experimental error is supposed to be the best design in many investigations.

Similarly, a design which yields maximal information and provides an opportunity for considering many different aspects of a problem is considered most appropriate and efficient design in respect of many research problems.

Features of a Good Design

An appropriate research design for a particular research problem usually involves the consideration of the following factors:

- the means of obtaining information
- the availability and skills of the researcher and his staff, if any
- the objectives of the problem to be studied
- the nature of the problem to be studied
- the availability of time and money for conducting a research

Developing a Research Plan

Arrangement of ideas in the form of an experimental plan is called a research plan. Research plan contains the following prerequisites:

- Research objectives should be clearly stated in line.
- Stated the problem explicitly so that one may know what information is to be obtained for solving the problem.
- Each major concept which a researcher wants to measure should be defined in operational terms in context of the research problem.
- The plan should contain the methods to be used in solving the problem.
- The plan must contain the details of the techniques to be adopted. For example, if interview method is used for data collection it should be written as such. Procedure for quantifying the data must also be given.
- Universe of the population, sampling plan, selection of appropriate sample size etc should be selected in such a way that all these are feasible.
- Processing of data should also be a part of the sampling plan.
- Results of the pilot test, if any, should be reported. Time and cost budgets for the research project should also be prepared and laid down in the plan.

Important Concepts Relating To Research Design

Before starting the research design, it would be appropriate to explain various concepts for easy understanding relating to design. It includes the following:

- Dependent and independent variables:
- Extraneous variable:
- Research hypothesis:
- Experimental and non-experimental testing of hypothesis:
- Experimental and Control groups:
- Treatments:
- Experiment:
- Experimental unit
- Sampling Unit

Research Hypothesis

- When a prediction or hypothesised relationship is to be tested by scientific methods, it is termed as research hypothesis. The research hypothesis is a predictive statement that relates a dependent variable to a dependent variable. Usually a research hypothesis must contain, at least, one independent & one dependent variable. Predictive statement which are not to be objectively verified or the relationships that are assumed but not to be tested, are not termed research hypothesis.
- **Experimental & control group:** In an experimental hypothesis-testing research when a group is exposed

- To usual conditions, it is termed a 'control group', but when a group is exposed to some novel or special condition, it is termed an 'experimental group'. For example, suppose a researcher randomly selects 50 students from a group of students who are to take a course in statistics & then divides them into 2 groups by randomly assigning 25 to group A, the usual studies program, & 25 to group B, the special studies program. Here, the group A can be called a control group & the group B an experimental group. If both groups A & B are exposed to special studies program, then both groups would be termed 'experimental groups'.

Research methods & research methodology

- It is important to explain the difference b/w research methods & research methodology. Research method may be understood as all those methods/techniques that are used for conduction of research. Research methods or techniques, thus, refer to the methods the researchers use in performing research operations. In other words, all those methods which are used by the researcher during the course of studying his research problem are termed as research methods.
- **RESEARCH METHODOLOGY:** Research methodology is a way to systematically solve the research problem. It may be understood as a science of studying how

- research is done scientifically. In it we study the various steps that are generally adopted by a researcher in studying his research problem along with the logic behind them. It is necessary for the researcher to know not only the research methods/techniques but also the methodology.
- We can say that research methodology has many dimensions & research methods do constitute a part of the research methodology. The scope of research methodology is wider than that of research methods. Thus, when we talk of research methodology we not only talk of research methods but also consider the logic behind the methods we use in the context of our study.

Quantitative & Qualitative designs

- The quantitative-qualitative classification is dependent on 3 criteria:
- (i) the purpose of study, (ii) How the variables are measured; & (iii) how the information is analysed.
- The study is classified as qualitative if: the purpose of the study is primarily to describe a situation, pheno-menon, problem or event; the information is gathered through the use of variables measured on nominal or ordinal scales (qualitative measurement scales); & if analysis is done to establish the variation in the situation, phenomenon or problem without quantifying it. The description of an observed

- situation, the historical enumeration of events, an account of the different opinions people have about an issue, & a description of living conditions of a community, are examples of qualitative research.
- On the other hand, if we want to quantify the variation in a phenomenon, situation, problem or issue, if information is gathered using predominantly quantitative variables, & if the analysis is geared to ascertain the magnitude of the variation, the study is classified as quantitative study. Examples of quantitative aspects of a research study are: How many people have a particular problem? How many people hold a particular attitude?

Types of Research Design

- **Two-sample t-test:**

Comparison of two means from independent populations

- **Paired-samples t-test for dependent samples:**

Information/data are collected from the same individual of a population i.e. before training and after training etc.

- **Chi-square test for association:**

To develop a relationship between two or more qualitative variables.

- **Regression and Correlation techniques:**

To establish a relationship between dependent and independent variable(s) (Regression); to establish a relationship between random variables in terms their degree of measurements (the variables under study are quantitative in nature).

Types of Research Design

- **One-way ANOVA or Completely Randomized Design (CRD):**

This design is used for comparison of more than two population means with the condition that all the units of an experiment are homogeneous.

- **Two-way ANOVA or Randomized Complete Block Design (RCBD):**

This design serves the same purpose like CRD but here the experimental units are grouped into units having similar characteristics and there exist one source of variation in the experimental units. For example, comparison of average monthly sales of stores located at different cities.

Types of Research Design

- **Three-way classification research design (Latin Square Design):**

If there exist two sources of variations in the experimental units then this design is preferred for mean comparison of different population (treatments/practices etc).

- **Factorial Designs:**

If multiple factors are involved in the experiment and one is interested to compare the main effects of the factors involved and their interactions, then factorial design is used for comparison.

- **Sample Design & Sample Survey:**
- **A** sample design is a definite plan for obtaining a sample from a given population. It refers to the techniques or the procedure (all principal steps) the researcher would adopt in selecting items for the sample. Sample design may as well lay down the number of items to be included in the sample, i.e. the size of the sample. Sample design is determined before data are collected. The term *survey* has been defined as a means of collecting information to meet a definite need. When a survey is carried out by a sampling method, it is called a *sample survey*. Researcher must select/prepare a sample design which should be reliable & appropriate for his research study.

- Steps in Sample Design:
- (i) Type of universe
- (ii) Sampling
- (iii) Sampling frame: A sampling frame is a complete list or a map that contains all the N sampling units in a population. A list of all the house-holds in a city, a map of a village showing all fields, etc. are examples of the frame.

Methods of Data Collection

- The task of data collection begins after a research problem has been defined & research design/plan chalked out. This means that when we have determined how to measure our variables, data w. r. t. each variable in the hypothesis need to be obtained. These data then form the basis for data analysis. While deciding about the method of data collection to be used for the study, researcher should keep in mind two types of data viz., primary & secondary. The primary data are those which are collected afresh & for the first time, & this happens to be original in character. The secondary data, on the other hand, are those which have already been collected by

- someone else & which have already been passed through the statistical process. The methods of collecting primary & secondary data differ since
- primary data are to be originally collected, while in case of secondary data the nature of data collection work is merely that of compilation.
- **Collection Of Primary Data:**
- We collect primary data during the course of doing experiments in an experimental research but in case we do research of the descriptive type & perform surveys, whether sample surveys or census surveys, then we can obtain primary data either through observation or through direct communication with respondents in one form or another through personal

- interviews. There are several methods of collecting primary data, particularly in surveys & descriptive researches. Important ones are:
- (i) Observation method (ii) Interview method (iii) Through questionnaire, (iv) through schedules.
- It may also be noted that sources of primary data are individuals, focus groups, panels of respondents specifically set up by researcher & from whom opinion may be sought on specific issues from time to time. Now we briefly take up the aforementioned methods separately.
- (i) Observation Method: Observation is a purposeful, systematic & selective way of watching & listening to

- an interaction, or phenomenon as it takes place. There are many situations in which observation is the most appropriate method of data collection. Under the observation method, the information is sought by way of investigator's own direct observation with-out asking from the respondent. For example, in a study relating to consumer behavior, the investigator instead of asking the brand of wrist watch used by the respondent, may himself look at the watch, or recording traffic counts. Other examples are, when you want to study the dietary patterns of a populati-on , ascertain the functions performed by a worker, or study the behavior or personality traits of an individual. It is also appropriate in situations where

- full and/or accurate information cannot be elicited by questioning, b/c respondents either are not cooperative or are unaware of the answers. In summary, the observation method is the most commonly used method specially in studies relating to behavioral sciences.
- The researcher can play one of two roles while gathering field observational data-that of a nonparticipant observer or participant observer.
- Nonparticipant-observer:
- The researcher may act as a nonparticipant-observer by collecting the necessary data without becoming

- an integral part of the organizational system. For example, the researcher might sit in the corner of an office & watch & record how manager spends his/her time. Observers physically present-time consuming.
- Participant observer:
- Here, the researcher enters the organization or the research setting & becomes a part of the work team. For instance, if a researcher wants to study group dynamics in work organizations, then he/she may join the organization as an employee & observe the dynamics in groups while being a part of the work organization.

Structured & Unstructured Observational Studies

- Where the observer has a predetermined set of categories of activities or phenomenon to be studied, it is a structured observational study.
- On the other hand, at the beginning of a study, it is possible that the observer has no definite ideas regarding the particular aspects that need focus. Observing events as they occur may also be a part of the plan, as in many qualitative studies. In such cases , the observer will record practically everything that is observed. Such a study is an unstructured observational study. Unstructured observational studies are claimed to be the hallmark of qualitative research.

- Some other methods of primary data collection include , consumer panels; using mechanical devices; through projective techniques; and depth interviews etc.

Interview Method

- The interview method of collecting data involves presentation of oral-verbal stimuli & reply in terms of oral-verbal responses. This method can be used through personal interviews and, if possible, through telephone interviews.
- (a) Personal *interviews*: Personal interview method requires a person known as the interviewer asking questions generally in a face-to-face contact to the other person or persons. At times the interviewee may also ask certain questions & the interviewer re-sponds to these, but usually the interviewer initiates the interview & collects the information. This sort of interview may be in the form of direct personal

- investigation or it may be direct oral investigation.
- *Structured & Unstructured Interviews:*
- The method of collecting information through personal interviews is usually carried out in a structured way. As such we call the interviews as *structured interviews*. Such interviews involve the use of a set of predetermined questions & of highly standardised techniques of recording. As against it, the *unstructured interviews* are characterised by a flexibility of approach to questioning. Unstructured interviews do not follow a system of pre-determined questions & standardised techniques of recording information.
- (b) *Telephone interviews:* This method of collecting information consists in contacting respondents on

- telephone itself. It is not a very widely used method, but plays important part in industrial surveys , partic-ularly in developed regions.
- **Collection of Data through Questionnaires:**
- This method of data collection is quite popular, parti-cularly in case of big enquires. It is being adopted by private individuals, research workers, private & public organisations & even by governments. In this method a questionnaire is sent (usually by post) to the person concerned with a request to answer the questions & return the questionnaire. A questionna-ire consists of a number of questions printed or typed in a definite order on a form or set of forms.

- The method of collecting data by mailing the questionnaires to respondents is most extensively employed in various economic & business surveys.
- ***Main Aspects of a Questionnaire:***
- Quite often a questionnaire is considered as a heart of a survey operation. Hence it should be very carefully constructed. If it is not properly set up, then the survey is bound to fail. This fact requires us to study the main aspects of a questionnaire viz., the *General form, Question sequence & Question formulation & wording*.
- *1. General form:* So far as the general form of a questionnaire is concerned, it can either be structured or

- unstructured questionnaire. Structured questionnaires are those questionnaires in which there are definite, concrete & pre-determined questions. When these characteristics are not present in a questionnaire, it can be termed as unstructured or non-structured questionnaire.
- 2. *Question sequence*: In order to make the question-naire effective & to ensure quality to the replies rec-eived, a researcher should pay attention to the question-sequence in preparing the questionnaire. The opening questions should be such as to arouse human interest. Questions of a personal character or related to personal wealth should generally be avoid-ed as opening questions in a questionnaire.

- *3. Question formulation & wording:*
- With regard to this aspect of questionnaire, the researcher should note that each question must be very clear for any sort of misunderstanding can do irreparable harm to a survey. Questions should also be impartial in order not to give a biased picture of the true state of affairs.

Collection of Secondary data

- Secondary data means data that are already available i.e. they refer to the data which have already been collected & analysed by someone else. Secondary data are indispensable for most organisational research. There are several sources of secondary data. Some important sources are discussed below:
- (i) **Government & Semi-government publications:**
- There are many govt. & semi-govt. organisations that collect data on a regular basis in a variety of areas & publish it for use by members of the public & interest groups. Some common examples are: the census, vital statistics registration, labour force surveys, health reports, economic forecasts, demographic

- Information, Publications of the Statistics division, M/o Finance, Food & Agriculture, Industry, labour, the Federal & provincial Bureaus of Statistics, State Bank of Pakistan, Central cotton committee, stock exchange, Railway Board & District councils etc.
- (ii) **Research Organisations:**
- Many organisations collect & store a variety of data to support their operations: for example, payroll details, copies of letters, minutes of meetings & accounts of sales of goods or services. Consumer research organisations collect data that are used subsequently by different clients. Trade organisations collect data from their members on topics such as sales that are

- subsequently aggregated & published.
- (iii) Earlier research:
- For some topics, an enormous number of research studies that have already been done by others can provide you with the required information.
- (iv) Personal records:
- Some people write historical & personal records that may provide the information you need.
- (v) Mass media:
- Reports published in newspapers, magazines, books etc. may be another source of data. For example, quality daily newspapers contain a wealth of data, including reports about take over bids & companies share

- prices.
- (vi) Universities & other Institutions:
- Reports prepared by research scholars & economists etc. in universities & other such institutions in different fields of activities.
- Problems with using data from Secondary sources:
- When using data from secondary sources, the researcher must be careful as there may be certain problems with the availability, format & quality of data. The extent of these problems varies from source to source. While using such data, some issues you should keep in mind are:
- (i) Validity & reliability of data: The validity of information may vary markedly from source to source.

- The reliability can be tested by finding out such things about the said data: (a) Who collected the data? (b) What are the sources of data? (c) Were they collected by using proper methods? (d) At what times were they collected? (e) Was there any bias of the compiler? Note that the use of information from personal diaries, newspapers & magazines may have the problem of personal bias.
- (ii) Suitability of data:
- The data that are suitable for one enquiry may not necessarily be suitable in another enquiry. Hence, if the available data are found to be unsuitable, they should not be used by the researcher.

- (iii) Adequacy of data:
- If the level of accuracy achieved in data is found inadequate for the purpose of present enquiry, they will be considered as inadequate & should not be used by the researcher.
- (iv) Format:
- Before deciding to use data from secondary sources it is equally important to ascertain that the data are available in the required format. For example, you might need to analyze age in the categories 23-33, 34-48 etc., but in your source the age may be categorized differently, e.g., 21-24, 25-29, etc.

From all this we can say that the already available data should be used by the researcher only when he finds them reliable, suitable and adequate.

Exploratory Studies or Research

Exploratory research is research conducted for a problem that has not been clearly defined. It implies that not much is known about the situation at hand, or no information is available on how similar problems or research issues have been solved in the past. In such cases, extensive preliminary work needs to be done to gain familiarity with the phenomena in the situation and understand what is occurring, before we develop a model. Exploratory research is also

Known as formulative research. Exploratory research often relies on secondary research such as reviewing literature and/or data, or qualitative approaches such as informal decisions with consumers, employees, management or competitors, and more formal approaches through in-depth interviews, focus groups, projective method, case studies or pilot studies. Extensive interviews with many people might have to be undertaken to get a handle on the situation and understand the phenomenon.

Some qualitative studies (as opposed to quantitative data gathered through questionnaire, etc.) where data are collected through observation or interviews, are exploratory in nature. When the data reveals some pattern regarding the phenomenon of interest, theories are developed and hypothesis formulated for subsequent testing e.g., a company interviewed managers to explore the nature of managerial work. Based on the analysis of their interviewed data, they formulated theories of managerial roles, the nature and

types of managerial activities and so on.

Exploratory studies are also necessary when some facts are known, but more information is needed for developing a viable theoretical frame work. For instance when we want to get at the important factors that influence the development of women in organizations, previous studies might indicate that women are increasingly taking on qualities such as assertiveness, competitiveness & independence. The results of exploratory research are not usually

useful for decision making by themselves, but they can provide significant insight into a given situation. Although the results of qualitative research can give some indication as to the “why”, “how” and “when” something occurs, it cannot tell us “how often” or “how many”. Exploratory research is not typically generalizable to the population at large.

In sum, exploratory studies are important for obtaining a good grasp of the phenomenon of interest and advancing knowledge through

subsequent theory building and hypothesis testing. In simple words, the objective of exploratory research is to gather preliminary information that will help define problems & suggest hypotheses.

Explanatory Studies and Research

Studies that establish causal relationships b/w variables may be termed as explanatory research. Explanatory research is also known as casual research (cause and effect).

Explanatory research attempts to classify why and how there is a relationship b/w two aspects of a situation or phenomenon. This type of research attempts to explain, for example, why stressful living results in heart-attacks; why a decline in mortality is followed by a fertility decline; or

how the home environment affects children's level of academic achievement. Let us consider one other example, where you may find that a cursory analysis of quantitative data on manufacturing scrap rates shows a relationship b/w scrap rates and the age of machine being operated. you could go ahead and subject the data to statistical tests such as correlation in order to get clearer view of relationship. Alternatively, or in addition, collect qualitative data to explain the reasons why customer of your company rarely pay their bills according to the prescribed payment terms.

Cross-sectional versus longitudinal studies

- Cross-sectional studies: A study can be undertaken in which data are collected just once, perhaps over a period of days, weeks or months, in order to answer a research question. Such studies are called one-shot or cross sectional studies. For example, a drug company desirous of investing in research for a new obesity pill , conducted a survey among obese people to see how many of them would be interest-ed in trying the new pill. This is a one-shot or cross-sectional study to assess the likely demand for the new product. The purpose of the study here is to collect data that would be pertinent to finding the

- answer to a research question. Data collection at one point in time is sufficient.
- Longitudinal studies:
- In some cases, the researcher might want to study people or phenomenon at more than one point in time in order to answer the research question. For instance, the researcher might want to study employees' behavior before & after a change in the top management, so as to know what effects the change accomplished. Here, b/c data are gathered at two different points in time, the study is not cross sectional, but is carried longitudinal across a period of time. Such studies, as when the data on dependent variable are gathered at two or more points in

- time to answer the research question, are called longitudinal studies.
- A cross sectional study is extremely simple in design. You decide what you want to find out, identify the study population, select a sample (if you need to), & contact your respondents to find out the required information.
- The incidence of HIV positive cases in a country.

- Collect data which have not been organized numerically are called *raw data*. For example, the weight of 120 randomly selected students (recorded to the nearest k.g) from a university is given as below.

67	63	57	85	67	60	75	55	67	68	51	54	45	57
64	68	67	86	63	60	98	83	76	70	56	50	74	74
67	77	61	85	66	66	60	61	58	56	56	57	60	60 ✓
63	64	85	80	75	75	57	58	59	58	58	61	62	91 ✓
74	72	57	73	61	86	64	91	64	64	61	62	69	57
81	66	65	81	82	76	77	81	76	66	62	63	62	63
60	60	72	72	79	70	70	58	78	58	71	76	60	60
65	65	66	65	73	73	71	73	66	73	67	68	69	68
73	68	74	68	67	76	52	79						

Main division of statistics

- (i) Descriptive Statistics (ii) Inferential Statistics
- (i) Descriptive statistics is that branch of statistics which deals with concepts & methods concerned with summarization & description of the important aspects of numerical data. It consists of *condensation of data*, their graphical displays, computation of numerical quantities such as specialized averages, ratios, projections & other measures which aid in making decisions.

- (ii) Inferential statistics or statistical inference comprising those methods concerned with the analysis of a subset (sample) of data leading to predictions or inferences about the entire set of data (population).
- It involves the estimation of population parameters and testing of statistical hypothesis. This area of statistics is based on probability theory, and lies in the heart of statistics. Note that larger group of data (or the whole) is called population and only a part of data is known as sample. These are defined below.

Population and Samples

- A population or statistical population is a collection or set of all possible observations whether finite or infinite, animate or inanimate, relevant to some characteristic of interest. In every day language, the word population means all the people in a specific area. However, the word population has a broader meaning in statistics. For example, population of bank accounts, all the houses in a city, all the students in a university, all the books in a library, all the patients in a hospital etc.

- Each individual item in a population is called an **element** of the corresponding population.
- Statistical population may be real such as the height of college students or hypothetical such as all the possible outcomes from the toss of a coin or throw of a dice.
- A sample is a part or subset of a population.
- No. of observations in a finite population is called the size of the population-denoted by N .
- It consists of some observations or whole of the population. The no. of observations in a sample is called the size of the sample is denoted by n .

- **PARAMETER & STATISTIC**

- A numerical quantity computed from a popn. (i.e. data obtained from a popn.) is called a parameter, and is denoted by Greek letter
- Values of parameters are generally unknown but may be estimated by using certain estimation method. For example.....
- A numerical quantity computed from a sample, is called a statistic---- represented by ordinary Latin letter.

- The information derived from sample data..... to draw conclusions about the popn. For example.....
- VARIABLE & CONSTANT
 - A characteristics that varies from individual to individual in a popn. For example, age is a variable as it varies from person to person. Similarly, other example are.....
 - Given set of all possible values from which the variable takes on value.... domain
 - Constant: If the domain of a variable contains only one value then the variable is referred Constant or fixed value

- **RANDOM VARIABLE**

- A variable whose value is obtained by chance---- or its value cannot be exactly determined. For example, suppose we toss a coin twice and call heads on a toss a success (0,1,2).
- Here, the occurrence of a particular value is a matter of chance, so the number of heads is a random variable. Similarly, the sum of dots obtained with a pair of die.....example of r.v.

- **QUANTITATIVE AND QUALITATIVE VARIABLE**

- Variables may be classified into quantitative and qualitative according to the form of characteristics of interest.

- Quantitative variable----- numerical value.
For example.....
- Qualitative variable-----not capable of taking numerical measurements. For example, education, intelligence.....A qualitative characteristic is also called an Attribute.
- Another definition: A characteristic which varies only in magnitude from one individual to another is called a quantitative variable.
- A characteristic which varies only in quality from one individual to another is called a qualitative variable or an attribute.

No. of outcomes when a pair of die is rolled

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Discrete and continuous variables

- A discrete variable is one that can take only obtained by a discrete set of integers or whole numbers, that is the values are taken by jumps. It is obtained by the operation of counting. For example, the No. of books and journals in a library,.....
- A continuous variable is one that take on any value-fractional or integral-within a given interval. A continuous variable is obtained by the process of measurement. For example, speed

- of car, the temperature of Islamabad etc.
- **Experiment and Event**
- The term experiment means a planned activity or process whose results yield a set of data. A single performance of an experiment is called a trial. An outcome of an experiment is called an event. An experiment which produces different results even though it is repeated a large number of times under essentially similar conditions, is called a random experiment. For example, tossing of a fair coin, drawing of a card from a deck of 52 playing cards, throwing a dice etc

Probability

- A probability is a number which means relative frequency of occurrence over the long run or expresses a degree of belief. In common language, we use the word chance rather than probability.
- When we say the probability that a tossed coin will land head up is 50% (or 0.5), we are using the relative frequency meaning of probability.
- Probability always lies between 0 and 1 (non-negative). Note 0 means 0% & 1 means 100%.

Simple Random Sample

- A simple random sample is one selected in a manner such that each element of the population has an equal chance of being included in the sample.
- Methods of selection of a S.R.S. Lottery system/draw, computer etc.
- S.R. Samples are important in the development of probability and statistical inference. Simple random sample & random sample are same.

Rounding numbers

- The process of rounding off or simply rounding a number means that a certain number of digits counted from the left, are to be retained and the last few digits are to be (i) dropped in a decimal number or (ii) replaced with zeros in a whole number. We adopt “choose the closer” convention in rounding as explained below:
- If we wish to round 74.68 to three digits by dropping the 8, we may write the rounded number as either 74.6 or 74.7, we choose 74.7 because it is closer to 74.68.

- However, if 55.54 is to be rounded to three digits, we write 55.5 rather than 55.6, why?
- When the digit to be dropped is exactly 5, or 5 followed only by zeros, the accepted practice is to increase the last significant digit by one if it is odd, and to leave unaltered if it is even. For example, the numbers 4.535 and 76.85 are rounded off 3 significant digits as 4.54 and 76.8, respectively. Think why ?
- Question: How 34.15 will be rounded off to three significant digits ? 34.2
- How 2.145001, 5.3772, 2.1548 and 46.528 are rounded to three significant digits ?

Percent change or difference

- A numerical change is a change from one number to another. An increase is change from the smaller of two numbers to larger number. Similarly a decrease is a change from the larger..
- A percent increase is computed by dividing the difference in two numbers by the smaller one. But a percent decrease is computed by dividing the difference by the larger of two numbers.
Example: During 21 working days this month, output of a plant was 6867 units. Last month had

- 24 working days and output was 6984 units. Compute percent change in output per working day.

Percentage Points

- The term percentage points is used to describe the numerical difference between two percent's. For example, if the output percent of radios in period I is increased from 111.1 percent to 136.1 percent in period II, then this is an increase of $136.1 - 111.1 = 25$ percentage points. Note, however, that the percent increase is $25/111.1 = 0.225$ or 22.5 percent.
- The percentage points is used to distinguish between the difference of two percent numbers.

Percent Error

- The difference between an approximation of a number and the correct number (i.e. approximation minus correct) is called the error.
- Percent error is computed by dividing the error by the correct number and expressing the result as a percent. For example, suppose an approximate record indicates that there are 156 calculators in stock but an actual count shows the correct number is 150. The error is $156 - 150$, or 6. The percent error is $\text{Error} / \text{correct number}$

- $=6/150= 0.04$ or $0.04 \times 100 = 4$ percent. It implies that the approximate inventory record is 4 percent greater than the correct count, or it can be said that, the record is in error by 4 percent on the high side.
- Next, suppose that the approximate inventory record had indicated 144 calculators. Then the error is $144-150= -6$; i.e. the record is 6 below the correct count. The percent error would have been $-6/150= -0.04$ or $-0.04 \times 100 = -4$ percent or 4 percent on the below side.

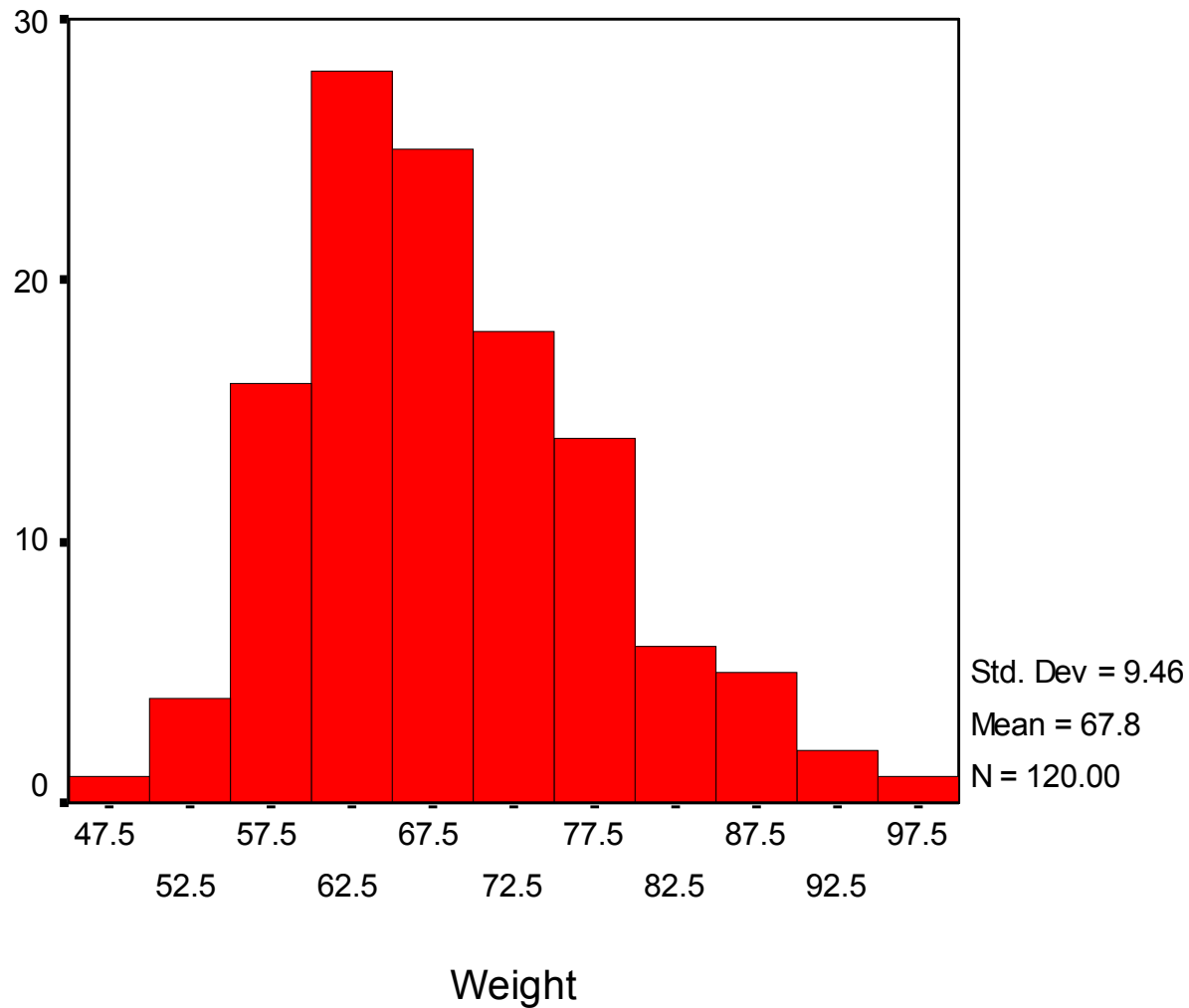
Percentage distribution

- If each of the numbers in a group is expressed as a percent of the sum of the group, the results are called a percentage distribution. For example, in the following table, sales by territory are shown in the top row. Their sum is 3185.
- Sales in City

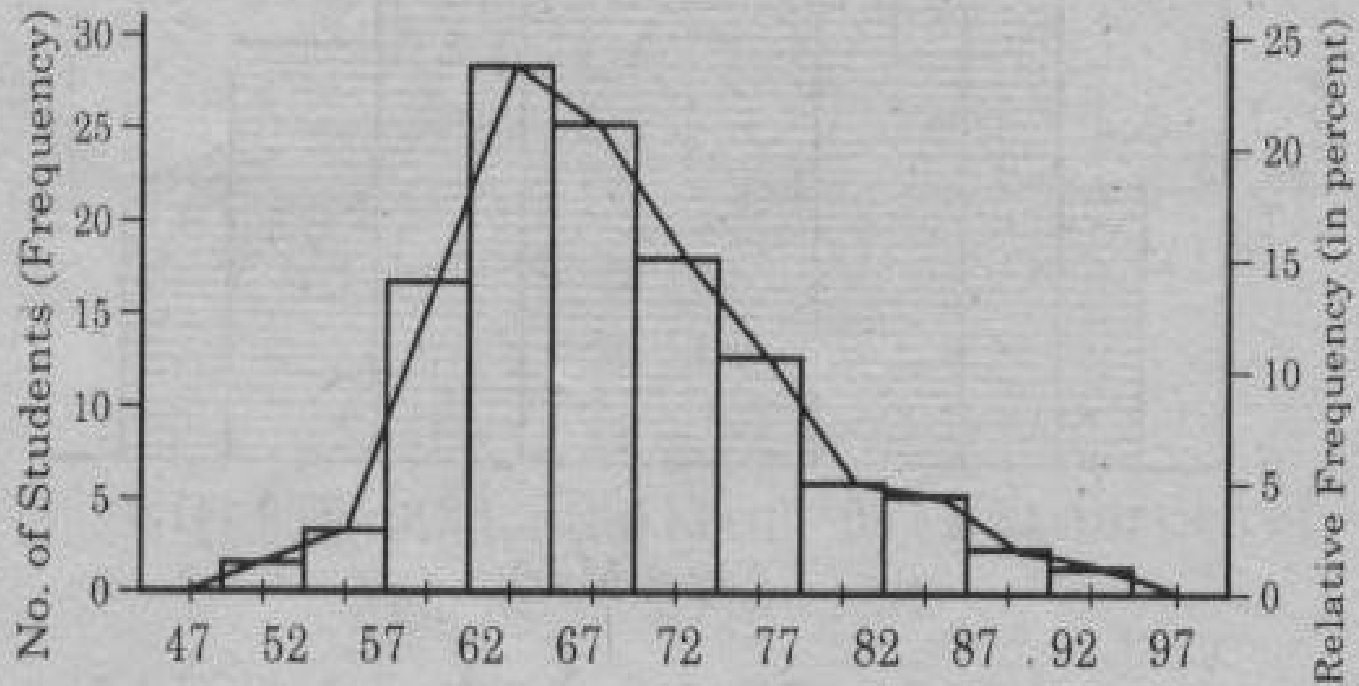
Summary/Descriptive Statistics of weight data of 120 students:

- Minimum value =45 kg
- Maximum value=98 kg
- Range=53 kg
- Arithmetic mean=67.83 kg
- Median=66.5 kg
- Mode=63.43 kg
- Q1=60.93 kg
- Q3=73.67 kg
- Standard deviation=9.52 kg
- Variance=90.63 kg

HISTOGRAM OF WEIGHT OF 120 STUDENTS

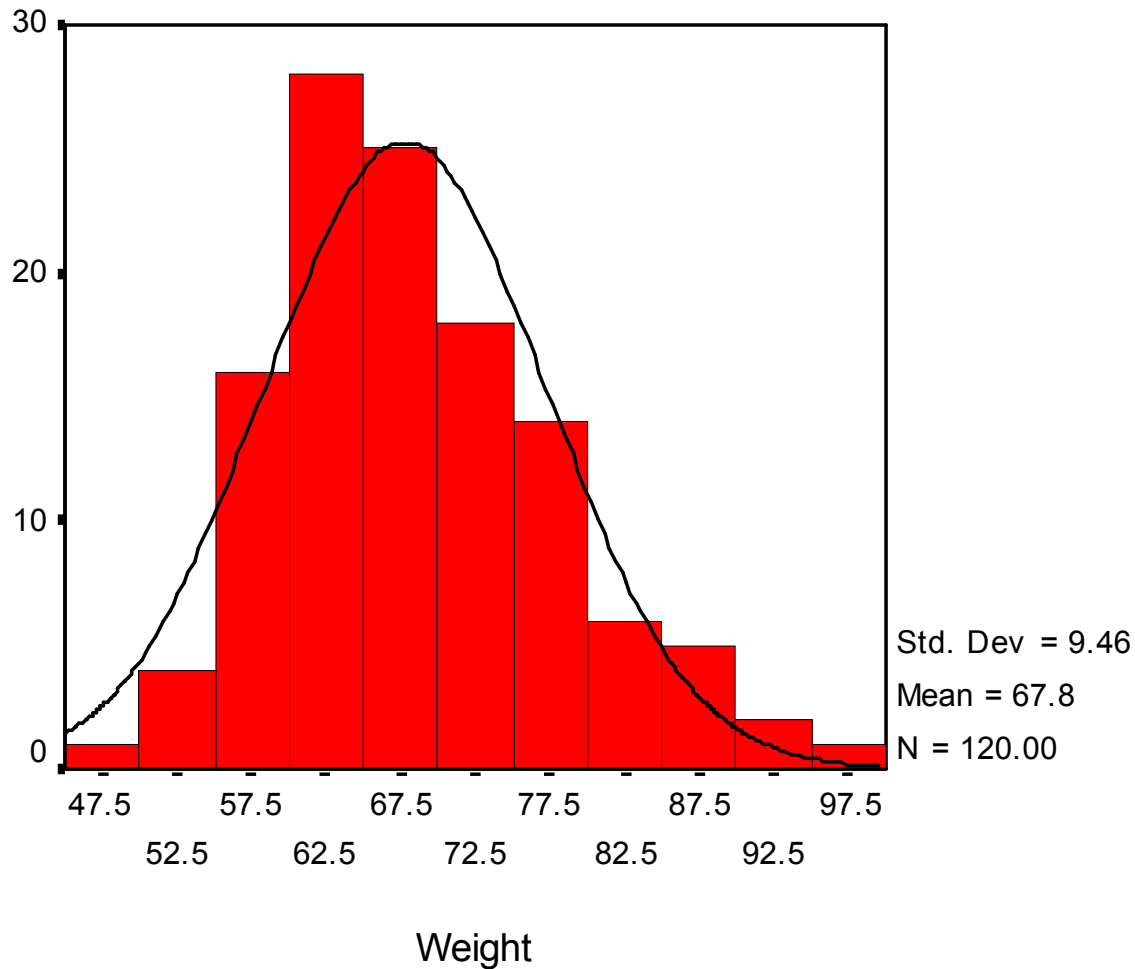


Histogram/Frequency polygon

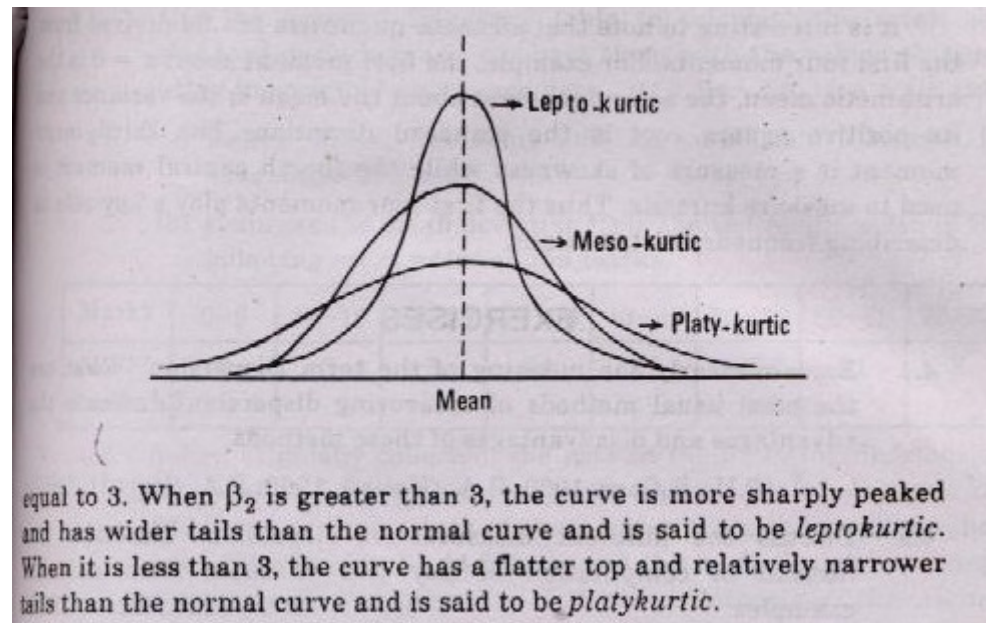


Frequency Polygon for the Frequency Distribution
of Weights of 120 Students

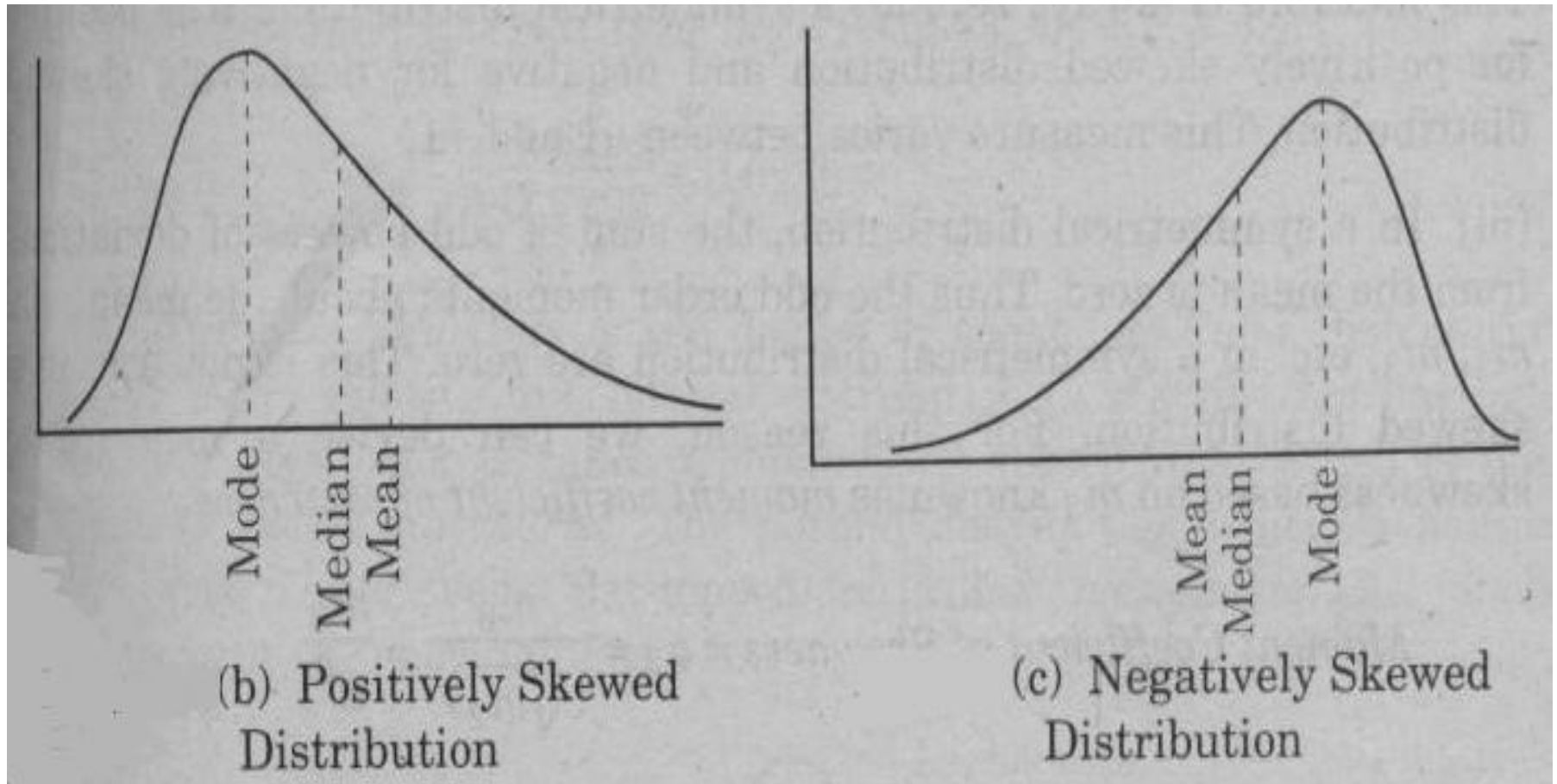
HISTOGRAM AND NORMAL CURVE OF WEIGHT OF 120 STUDENTS



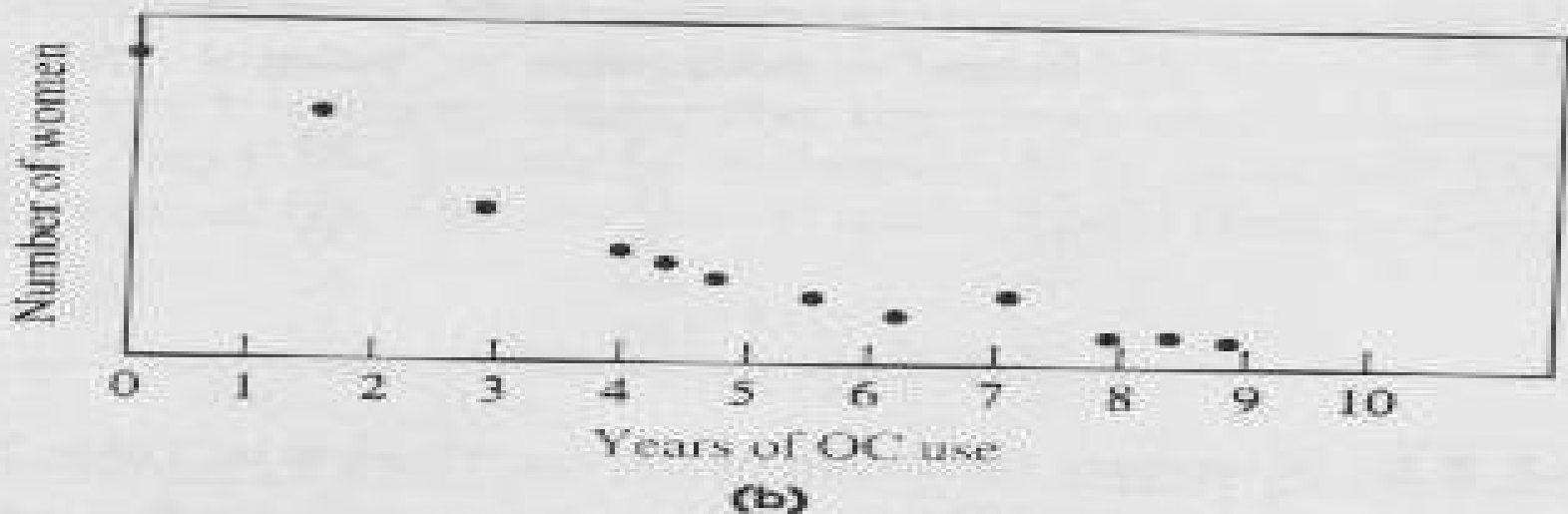
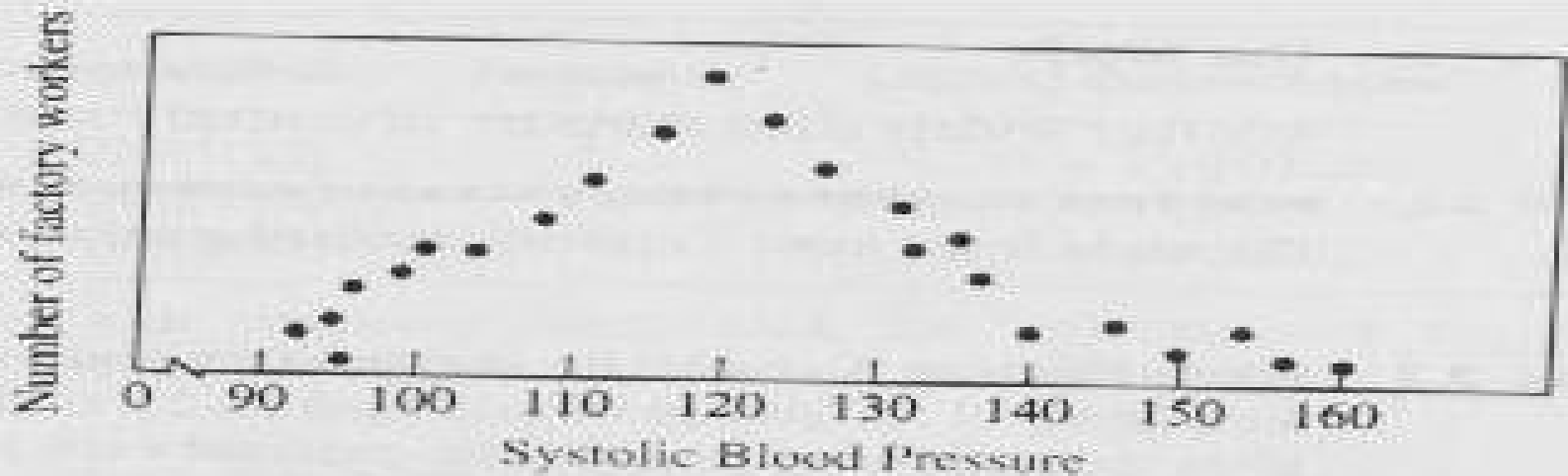
Normal distribution



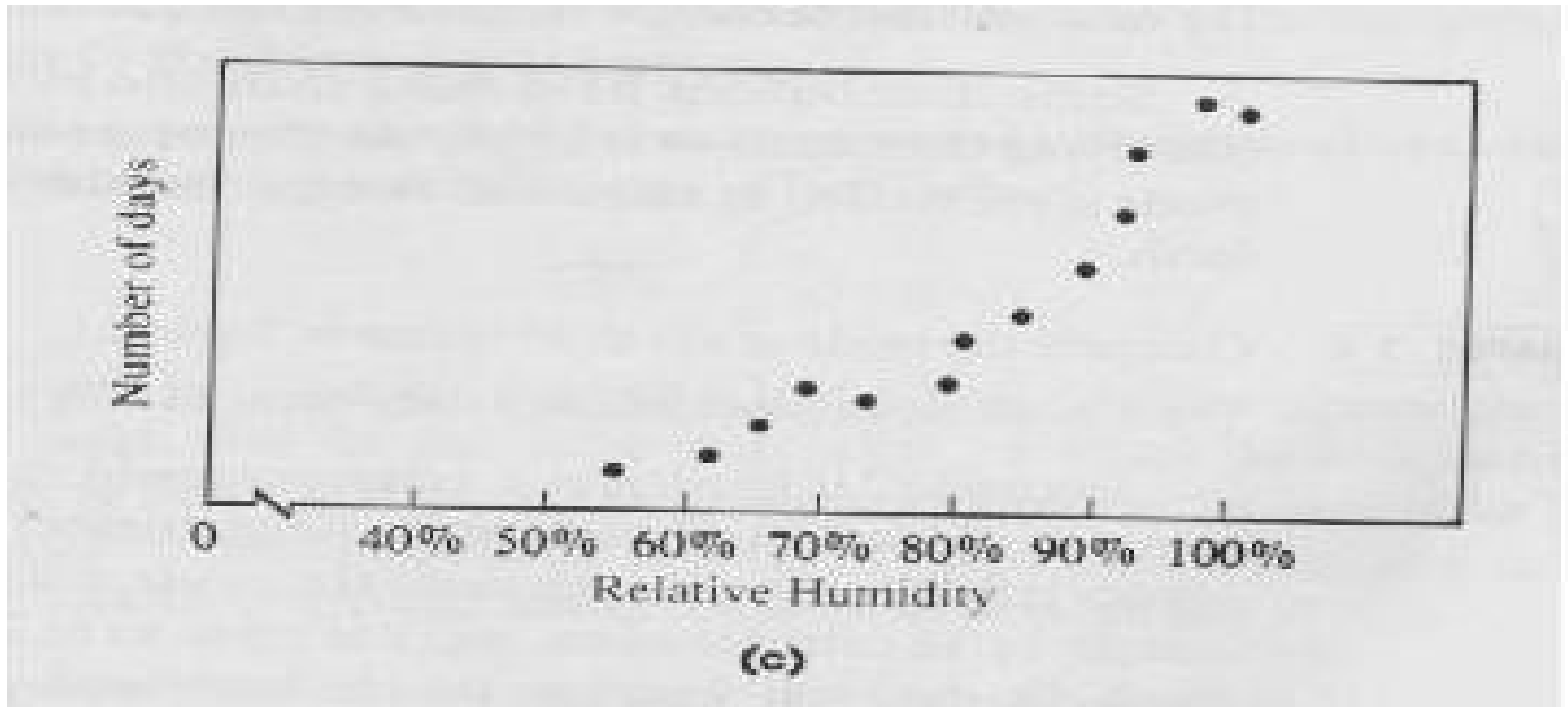
Skewness (degree of asymmetry)



Graphical display of skewness (symmetrical & positively skewed)



Graphical display of skewness (negatively skewed)



For Normal distribution:

- a) The interval $\bar{x} - \sigma$ & $\bar{x} + \sigma$ includes 68.27% of the cases/observations. The interval $\bar{x} - 2\sigma$ & $\bar{x} + 2\sigma$ includes 95.45% of the cases/observations.
 - b) The interval $\bar{x} - 3\sigma$ & $\bar{x} + 3\sigma$ includes 99.73% of the cases/observations.
- Note:- For moderately skewed distribution the above percentages hold approximately.

Scale of measurements

- (1) **Nominal**:-

Numbers represent categories. Numbers do not reflect differences in magnitude. Numbers serve to distinguish groups.

Example:- Sex, nationality, clinical diagnosis place of domicile.

- (2) **Ordinal**:-

Numbers indicate rank order of observations.

Examples:- Social class, level of education, performance evaluation.

- (3) **Interval**:-

Numbers describe the magnitude of the differences among the things. Here, the zero point on the scale is arbitrary and does not correspond to the absence of the trait/characteristic being measured.

Example:- Temperature, calendar year (A.D.)

- **Ratio**:-

Numbers represent equal units from absolute zero. Observations can be compared as ratio or percentages.

Examples:- Distance, age, time, weight, blood pressure.

Hypothesis Testing (6 steps)

- Statement of null & alternative hypothesis.
- Decide upon level of significance, α .
- Test-statistic to be used (decision).
- Critical region or region of rejection.
- Computation/Calculation.
- Result/interpretation.

Independent & dependent (Paired) t-test (Testing hypothesis about two means)

- Example: The systolic blood pressure of 10 women were recorded before and after administering the drug (calcium channel blocker). Test the hypothesis (at the 5% level) That the drug affects the average blood pressure Of women.
- 2 sample t- test is used to test the equality of two population mean, i.e. $H_0: \mu_1 = \mu_2$ and $H_1: \mu_1 \neq \mu_2$

B.P. before taking drug (mm/Hg.)	B.P. after taking drug (mm/Hg.)	Difference d_i
128	115	13
115	112	3
106	107	-1
128	119	9
122	115	7
145	138	7
132	126	6
109	105	4
102	104	-2
117	115	2

- $t = \frac{\bar{d}}{s_d/\sqrt{n}}$ where $d=4.80$ ____
 $n=10$

& standard deviation of difference $=s_d= 4.566$,
 $t=(4.80)/4.566 \times 3.162= 3.32$
 $t_{\text{tab}} = t_{0.025(9)} = 2.262$

Result:- H_0 is rejected which implies that drug has significant effect in reducing the B.P. of women.

ANOVA TEST: ANOVA technique is used to test the equality of more than two population means, i.e.: $H_0: \mu_1 = \mu_2 = \dots = \mu_k$.
 For this purpose, F-test/distribution is applied.

- Ten students were subjected to physical training. Their body weights before and after the training were recorded as given below. Test whether there is any significant effect of training on body weight or not.

Weight	1	2	3	4	5	6	7	8	9	10
Before	55	48	45	62	48	59	37	40	48	52
After	56	51	48	60	52	57	45	46	49	55

The following two varieties of wheat were planted in 8 different localities. The yields of these varieties (in K. g) are recorded as

Variety 1	25	15	36	34	40	18	27	31
Variety 2	28	14	40	39	43	24	29	33

Test the hypothesis to test the equality of mean yields for 2 varieties.

REGRESSION

- Introduced by English Biometrician, Sir Francis Galton in 19th century.
- Tendency to going back or regress towards average (height example).
- To establish functional/mathematical relationship b/w one dependent and one or more independent variables.

- Two main aims of regression are :
 - a) To establish relationship among variables.
 - b) Prediction of future values of the response variable.
- Deterministic and probabilistic models.
- Method of least squares (estimation of parameters).
- Estimating a & b using the formulae derived by MLS.

MLS

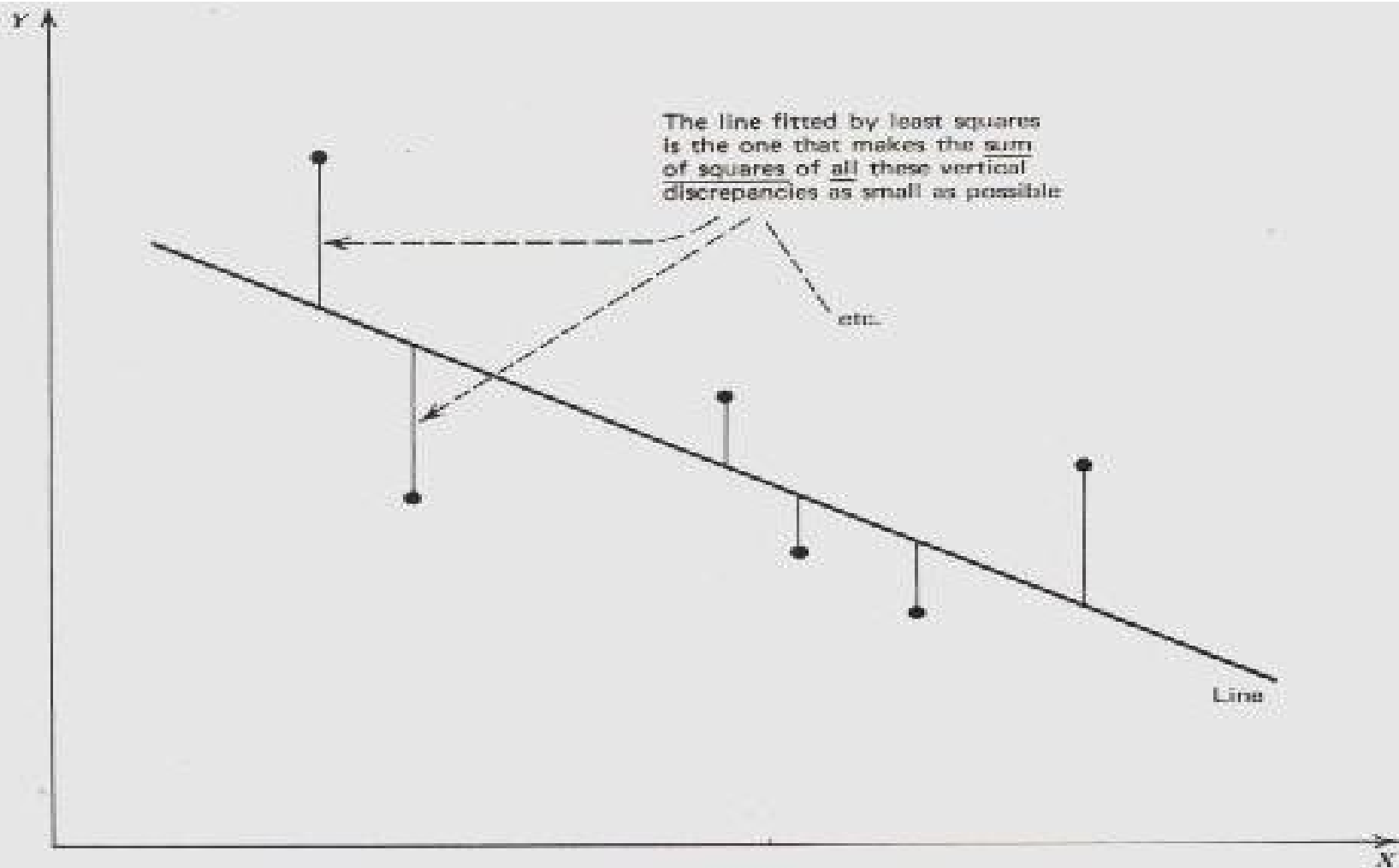


Figure 1.5 The vertical deviations whose sum of squares is minimized for the least squares procedure.

- General equation of linear regression

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon_i$$

($i=1,2,3,\dots,n$)

Where Y is dependent variable, $X_1, X_2, \dots, X_k$ are independent variables, α is intercept, $\beta_1, \beta_2, \dots, \beta_k$ are regression coefficients & ε_i 's are the random errors.

- Regression can be linear or non-linear or curve-linear.
- Simple and multiple linear regression.
- Simple correlation.
- Diagnostic checking of the fitted model or evaluating the Fit. It is very important to investigate the structure of the residuals and the data pattern through graphs.

Data of four graphs

SIMPLE LINEAR REGRESSION

Table Four data sets having same values of summary statistics

	X1	Y1	X2	Y2	X3	Y3	X4	Y4
001	10	8.04	10	9.14	10	7.46	8	6.58
002	8	6.95	8	8.14	8	6.77	8	5.76
003	13	7.58	13	8.74	13	12.74	8	7.71
004	9	8.81	9	8.77	9	7.11	8	8.84
005	11	8.33	11	9.26	11	7.81	8	8.47
006	14	9.96	14	8.10	14	8.84	8	7.04
007	6	7.24	6	6.13	6	6.08	8	5.25
008	4	4.26	4	3.10	4	5.39	19	12.50
009	12	10.84	12	9.13	12	8.15	8	5.56
010	7	4.82	7	7.26	7	6.42	8	7.91
011	5	5.68	5	4.74	5	5.73	8	6.89

Source: Anscombe (1973).

Four different graphs having same summary statistics

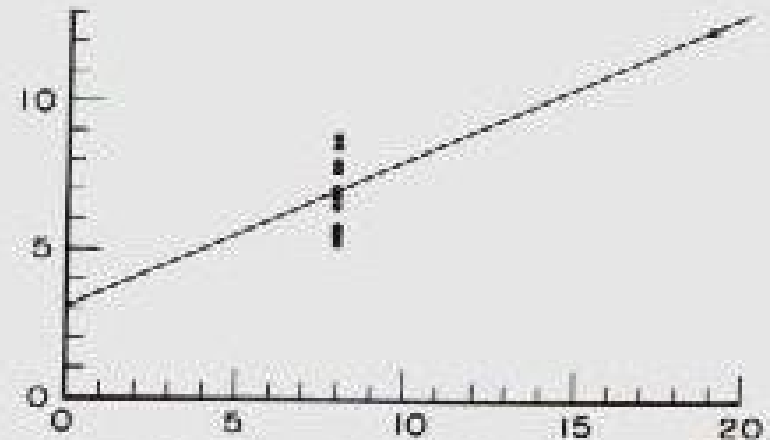
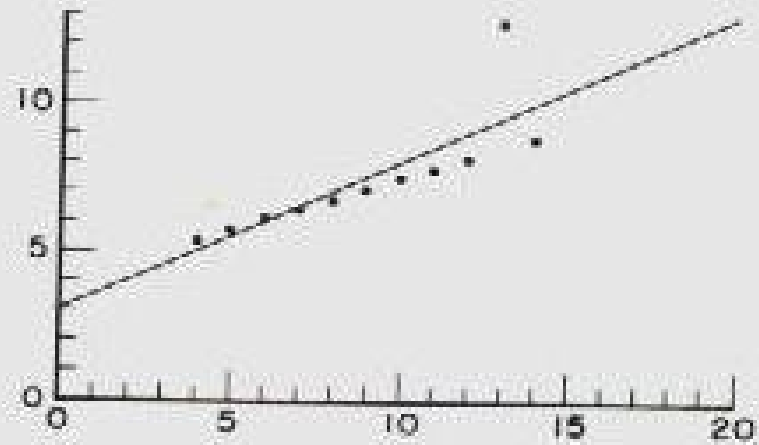
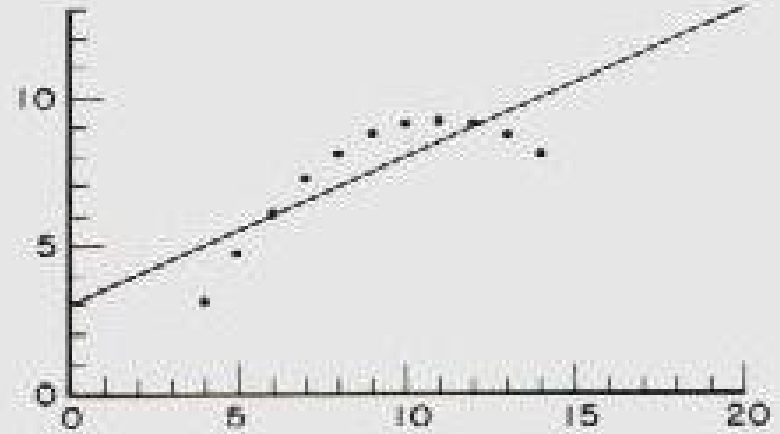
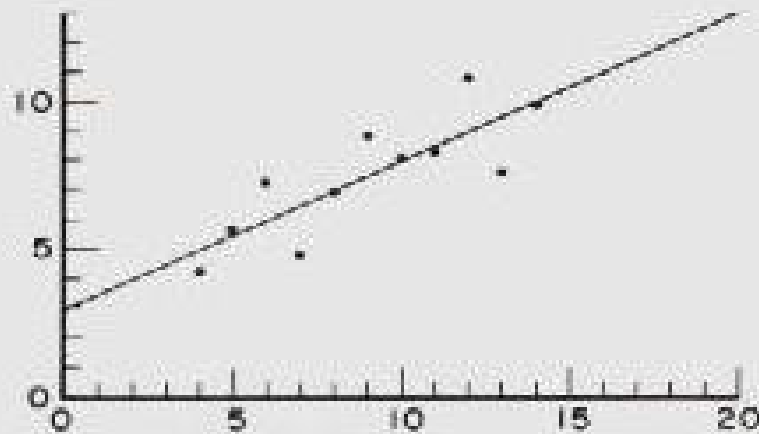
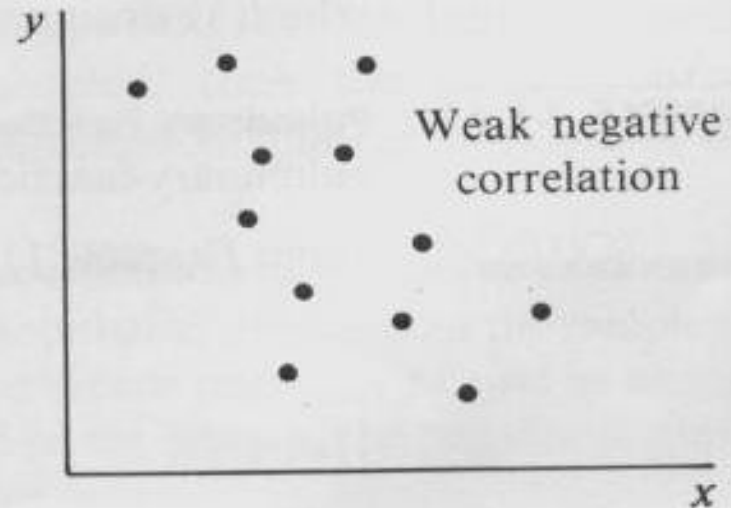
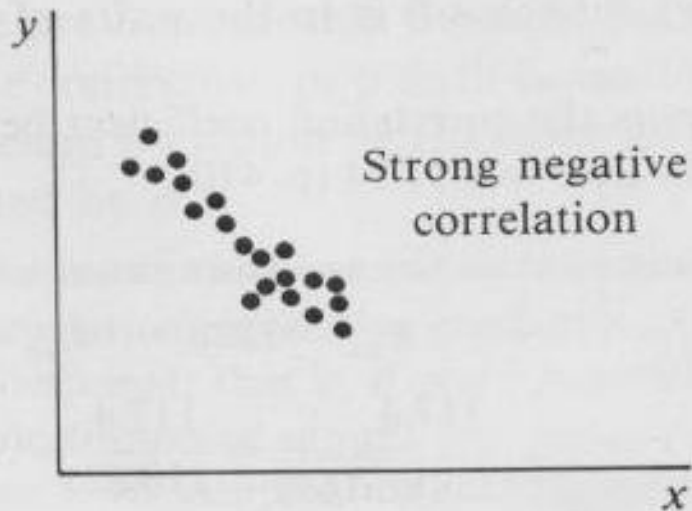
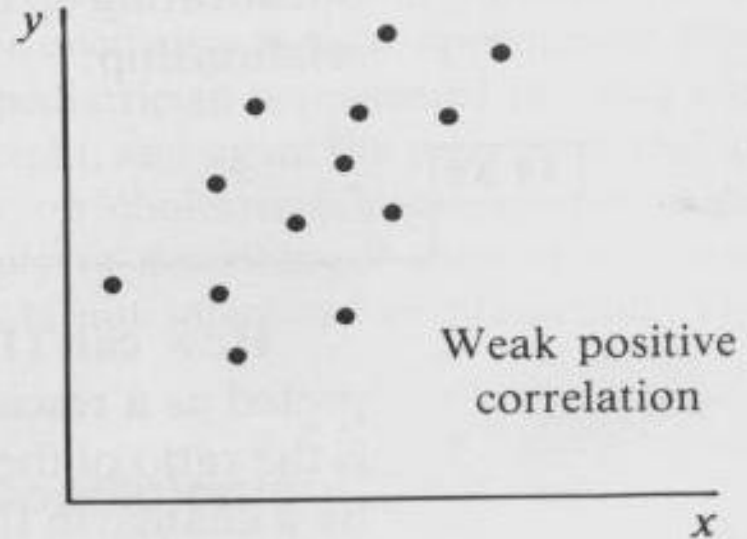
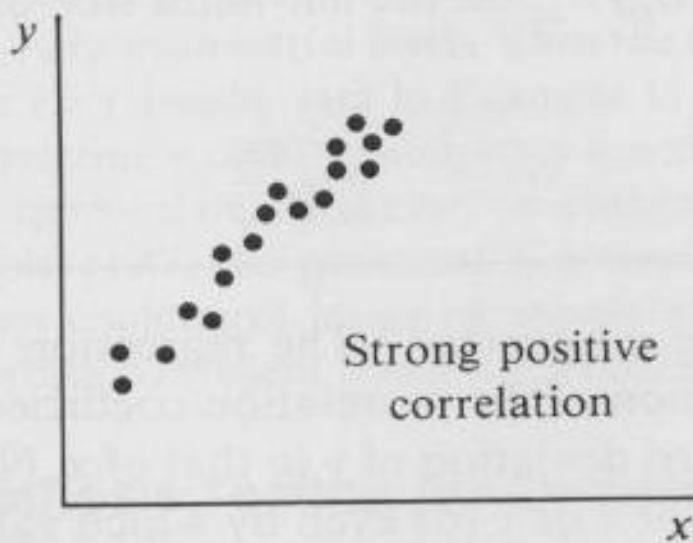
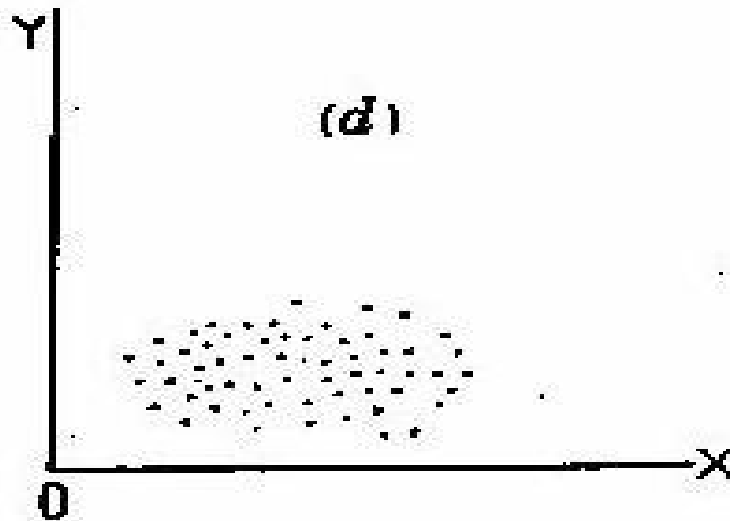
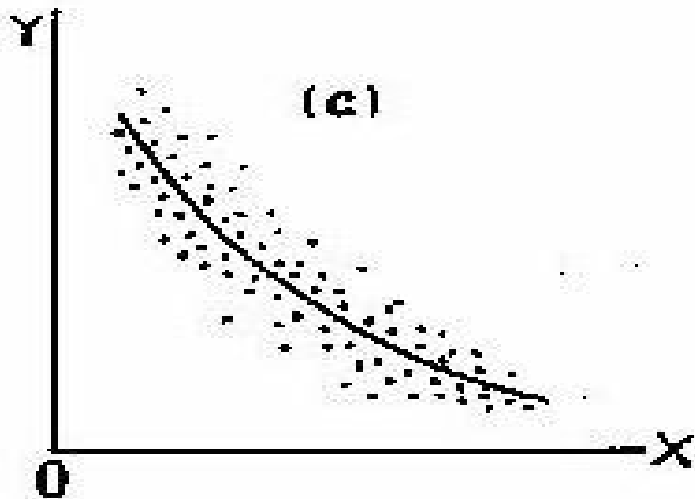
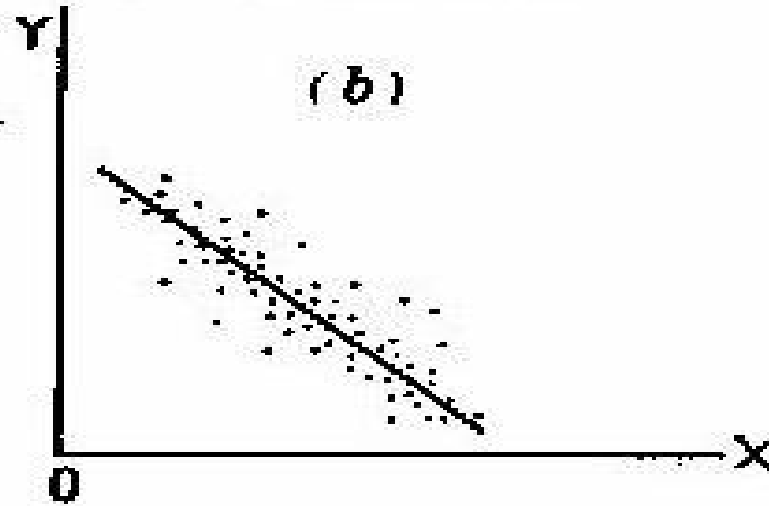
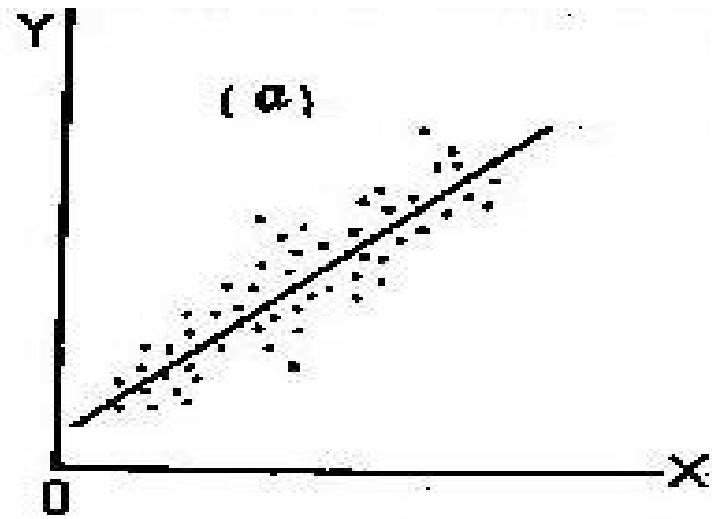


Fig. 1.1. Plot of the data (x,y) with the fitted line for four data sets (Table 1.1). Source: Anscombe (1973).

Interpretation of various degrees of correlation (scatter diagram)



Observed & fitted data

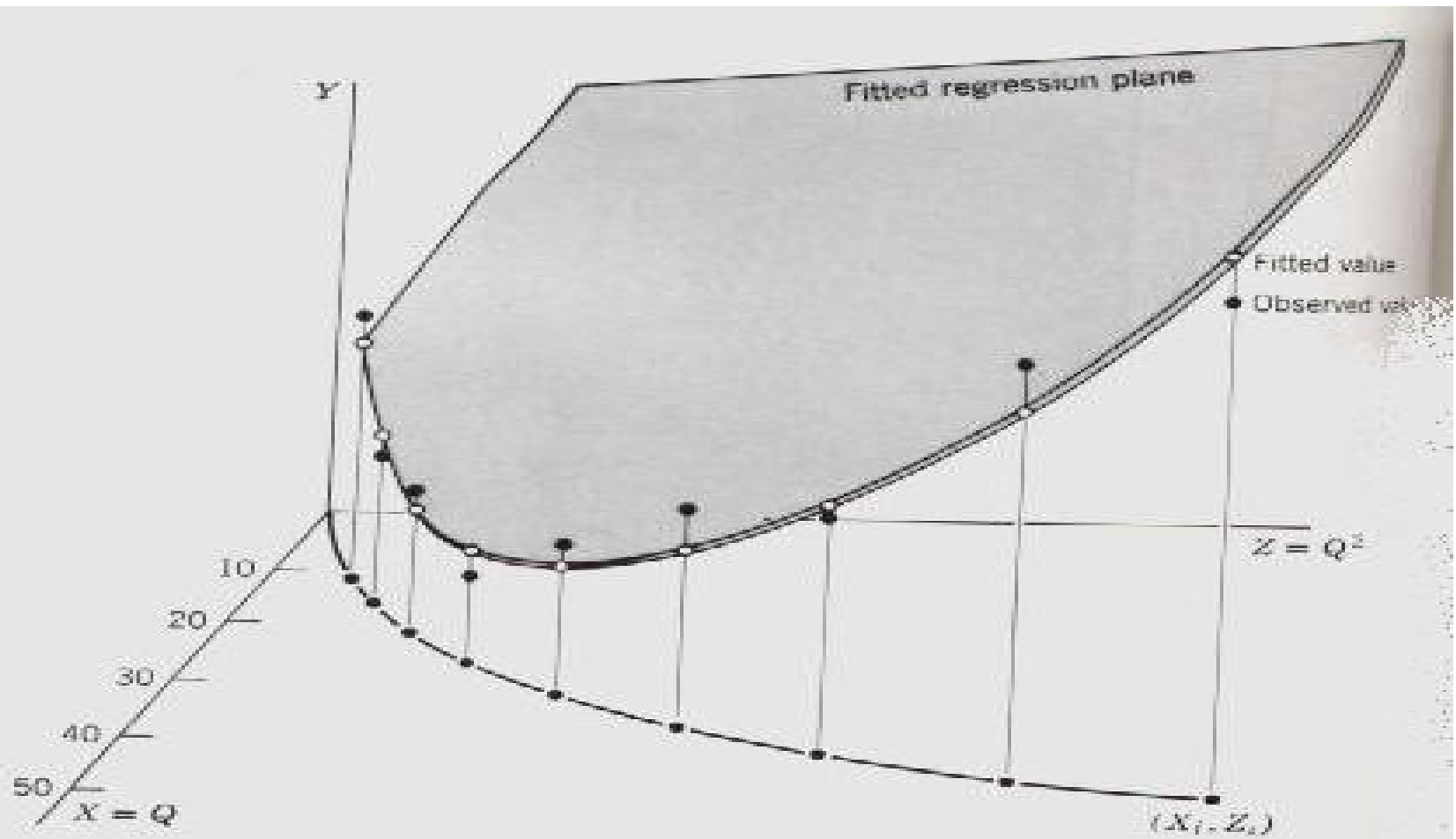


Multiple regression

- Usually simple regression is inadequate in real world situation
- Multiple regression is an extension of simple regression.
- $Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \varepsilon_i$
($i=1,2,3,\dots,n$).
- Estimate α , β_1 & β_2 using MLS.

.

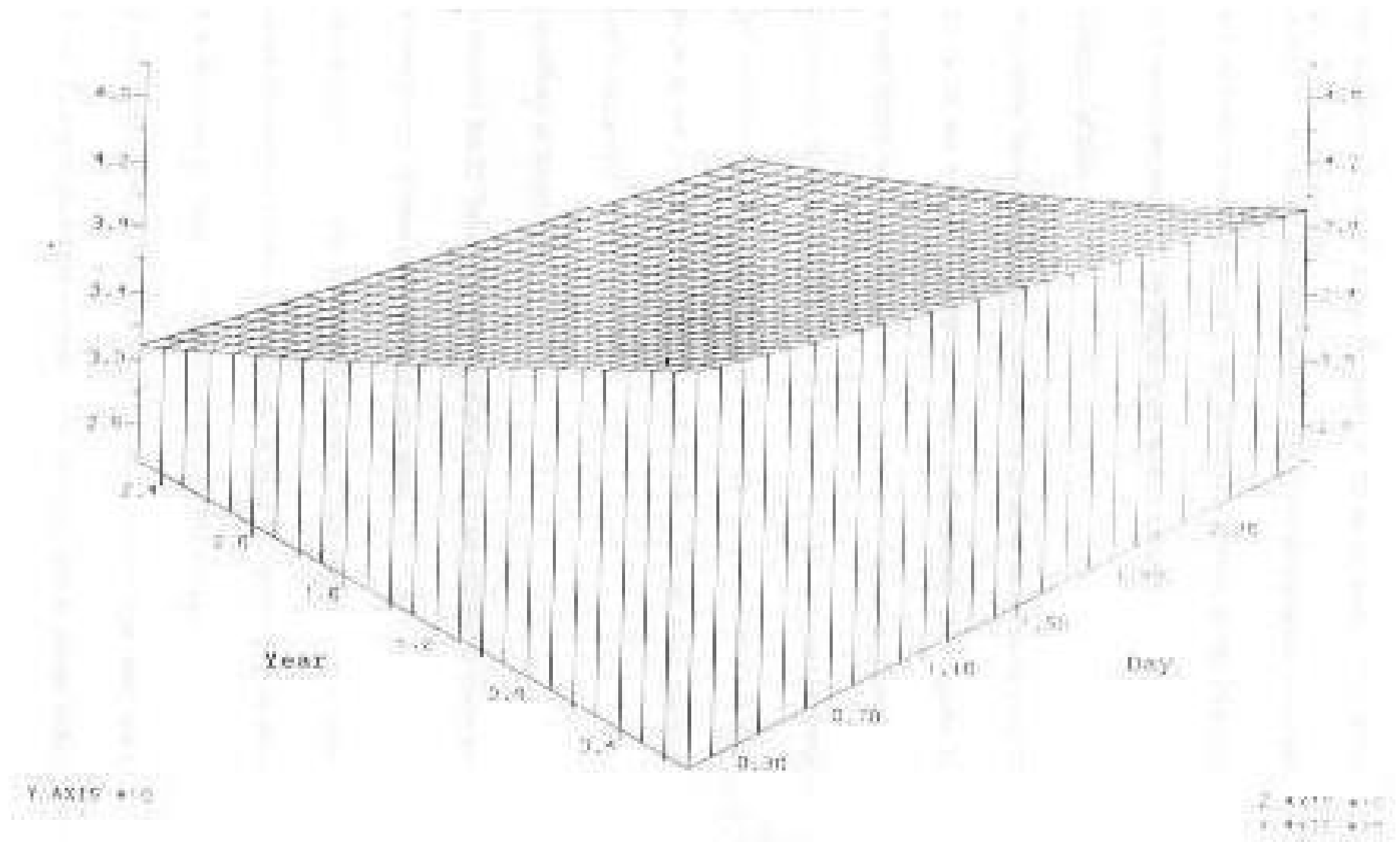
Multiple regression



FIGURE

Polynomial regression as a special case of multiple regression.

3 Dimension Graph 2



Common Violation in Agricultural Experiments

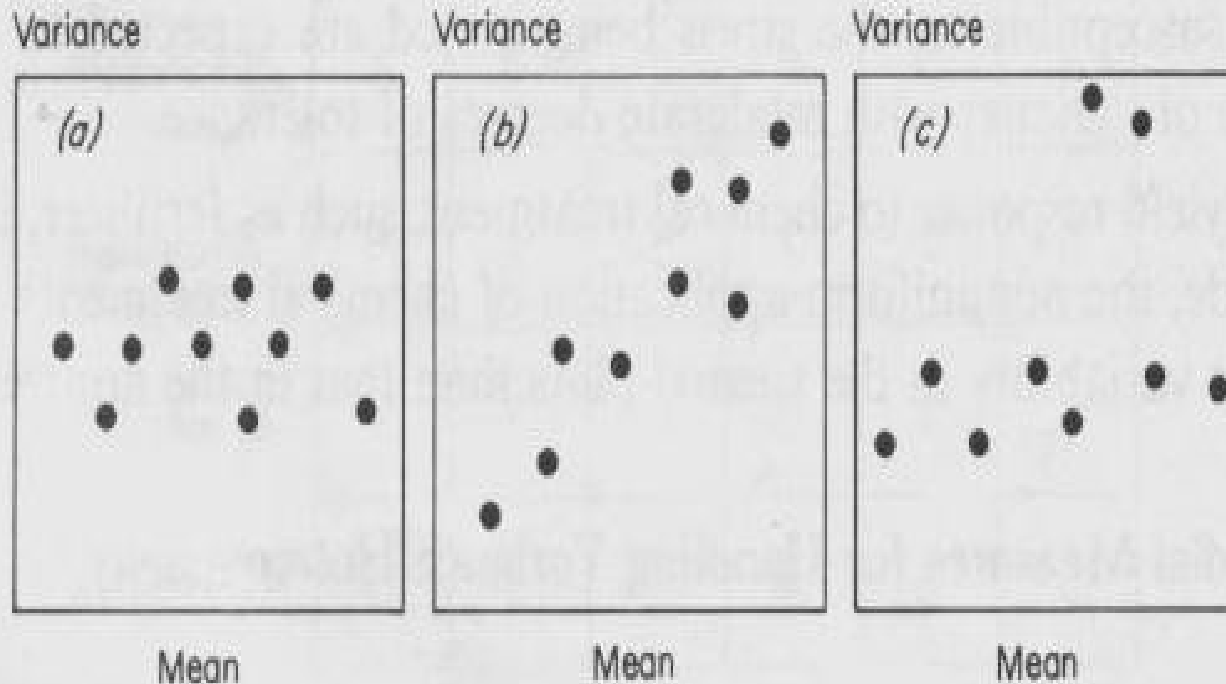
Variance heterogeneity and Nonnormality

- Where the variance is functionally related to the mean.
- Usually associated with data whose distribution is not normal.
- Examples (i) Such as the No. of infested plants per plot or the No. of lesion per leaf (usually follow a Poisson distribution where in $S^2=X$).
- (ii) Such as percent survival of insects or percent plants infected with a disease. Here each occurrence can only be one of the two possible outcomes (e.g. alive or dead and infested or not infested). these follow a binomial distribution.

Data transformation

- It is the most appropriate remedial measure for variance heterogeneity where the variance and mean are functionally related (see the 3 graphs).
- Original data are converted into a new scale resulting in a new data set that is expected to satisfy the condition of homogeneity of variance. A few data transformation techniques are described here.

Different types of variance heterogeneity.



(a) homogeneous variance,

(b) heterogeneous variance where variance is proportional to the mean, and (c) heterogeneous variance without any functional relationship between variance and mean.

Logarithmic transformation.

- This transformation is most appropriate for data where standard deviation is proportional to the mean. This condition is generally in data where numbers are large and cover a wide range of values. e.g. data on the number of insects per plot.
- To illustrate the procedure for applying logarithmic transformation, we use data on the number of living larvae on rice plants treated with various rates of an insecticide from a RCB experiment with four replications. The graphs are given below

- Some rules for choosing the proper transformation scale for %age data derived from count data are given here.

- **Rule1.**

For percentage data lying with the range of 30 to 70%, no transformation is needed.

- **Rule 2.**

For percentage data lying with in the range of either 0 to 30% or 70 to 100%, but not both, the square transformation should be used.

- **Rule 3.**

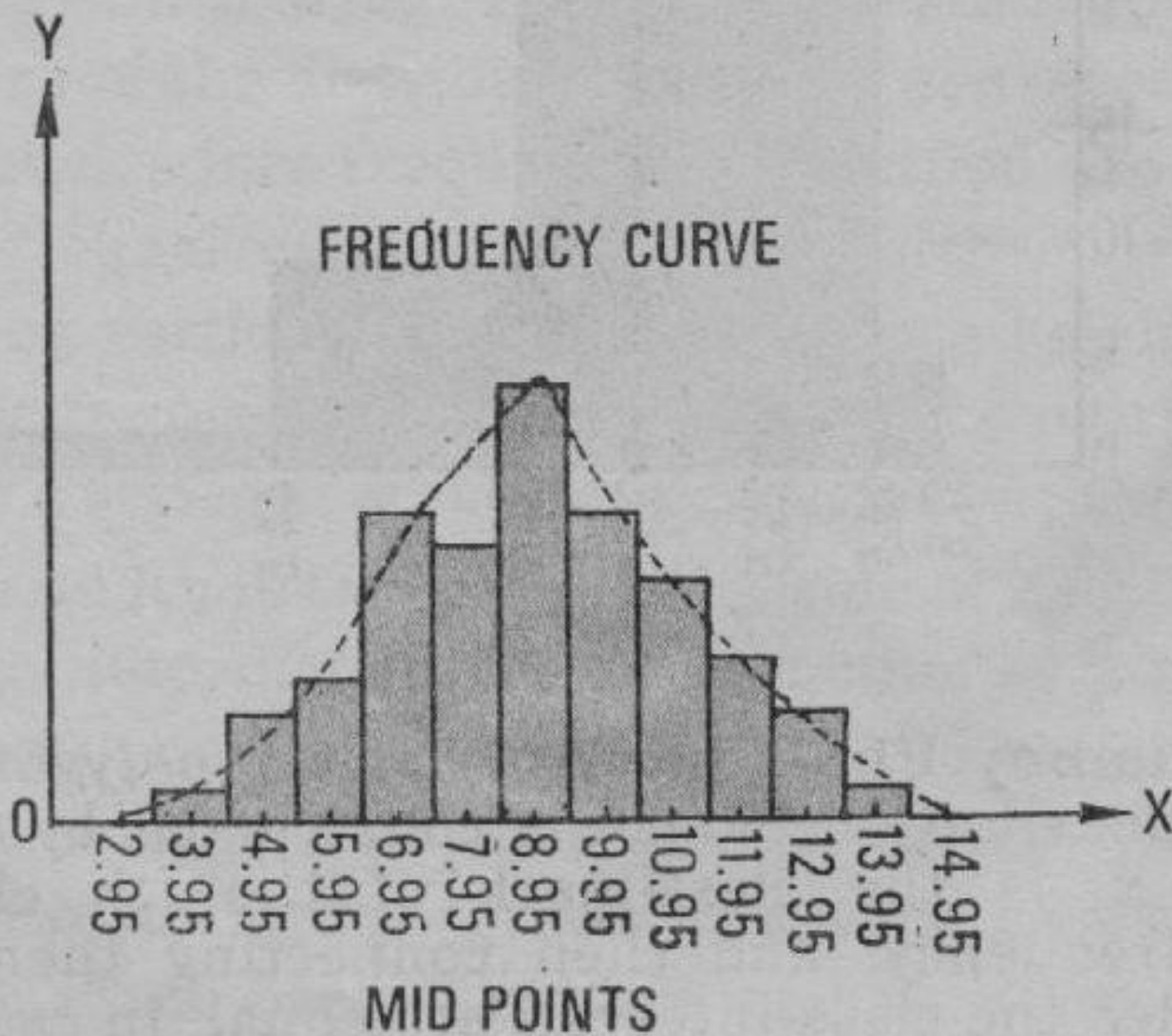
For percentage data that do not follow the ranges specified in either rule 1 or rule 2, the arc sine transformation should be used.

Suggested reading

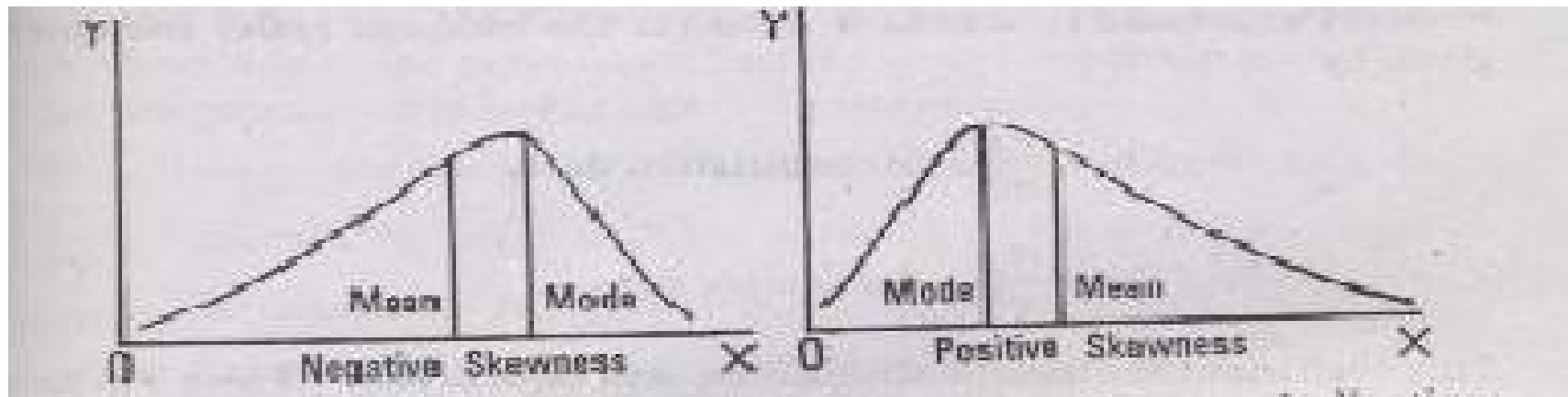
- Introduction to Statistical theory, part I & Part II (2001/latest edition).
by Prof. Sher Muhammad Chaudhry & Dr. Shahid Kamal.
- STATISTICS: Methods and Application (2005)
by Dr. M. Afzal beg
Karvan Book House, Lahore.
- Statistical Methods and Data Analysis (Edition:2000)
by Dr. Faqir Muhammad, Kitab Markaz Bhawana Bazar, Faisalabad.
- Agriculture & applied statistics-I by P.K Sahu (2007), Kalyani publishers.
- Bio statistical analysis 4th edition by J.H Zar, National book foundation, Islamabad
- Statistical procedures for agricultural research (1984/latest edition) by K.A Gomez & A.A Gomez.

THANK YOU

FOR YOUR ATTENTION



Skewness



Regression curves

Fig. B.7.1 Observed & fitted mean rainfall per rainy day at Seonasa

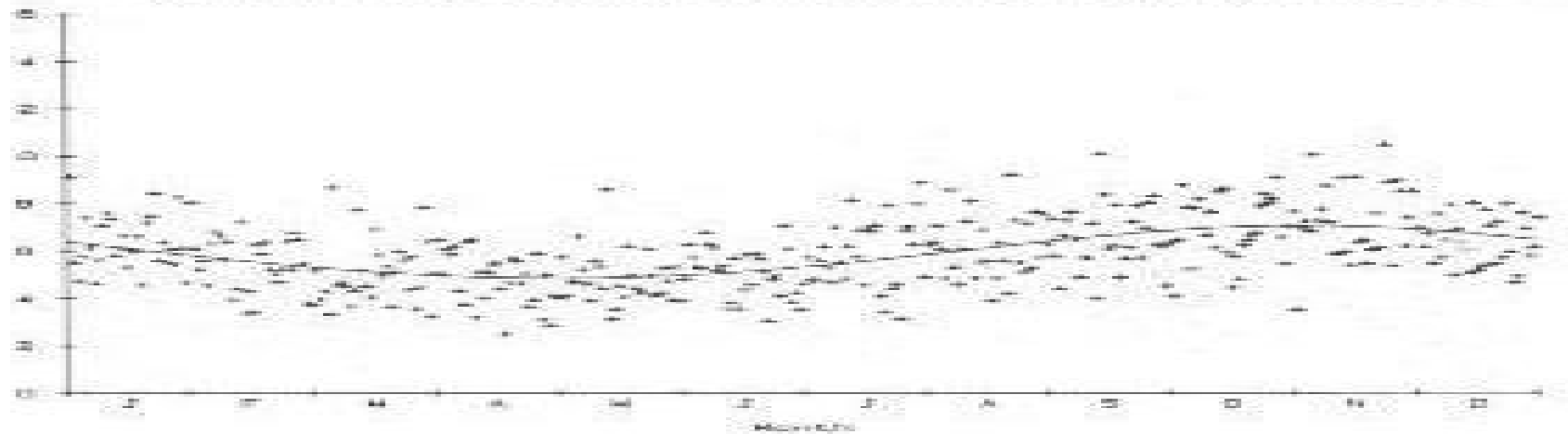
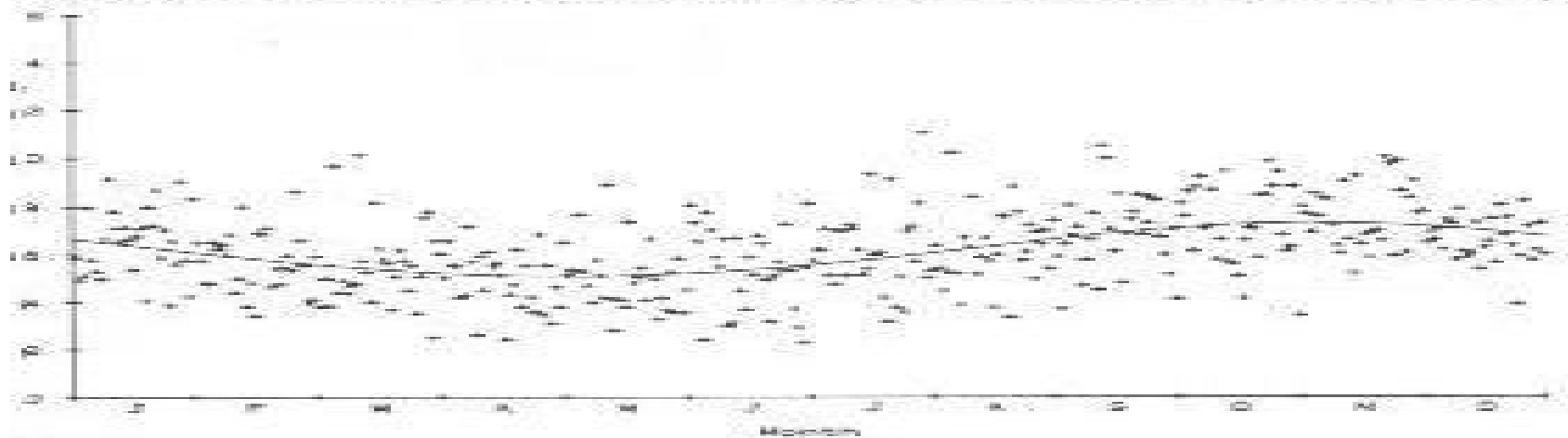


Fig. B.7.2 Observed & fitted mean rainfall per rainy day following a rainy day



Scale of measurements

- **Nominal measurement:**

A nominal scale is one in which it is possible to place the variable into unordered categories. For example, we may want to know how much patients in a hospital are hypertensive, diabetic manic-depressive & psychoneurotic, & so on. Data of this type are frequently referred to as count data.

Ordinal measurement:

- An ordinal measurement is one in which the results can be ordered along a given scale. Ordinal data are data that can be ordered but do not have specific numerical values. For example, students or employees can be graded according to their performance as 1=Excellent, 2=very good, 3= good, 4= satisfactory, 5= average. This is ordinal variable because the different grades 1,2,3,4,5 are ordered in the sense that grade 1 is better than 2, which is better than grade 3, -----, and so on. However, we cannot say that the difference between categories 1 and 2 is the same as the difference between grades 2 and 3, ----, and so on. This is also called Rank data.

Interval measurement.

- An interval measurement is one in which the increments are consistent across the entire measuring scale. However, it has the limitation that it does not include true zero point. The most common example is the temperature. An object can be exposed to 30° Fahrenheit, 60° F and 90° F. The actual difference between 30° and 60° is the same as between 60° and 90°, but there is no true zero point in the scale, it cannot be concluded that 60° represents twice as much heat as 30° or that 90° represents thrice as much heat as 30°. Such a conclusion would not be correct. It may be noted that in the example the zero point (that is arbitrary) has a different meaning for Fahrenheit & Celsius temperatures.

Ratio Measurement:

- The ratio measurement is one in which the intervals are consistent along the entire scale and there is a true zero point. For example, Blood pressure and body weight are on ratio scales (or measurement) because the zero-point is well-defined in both instances. In case of body weight, we can say that a difference of 6 lbs is actually twice as large as a difference of 3 lbs.

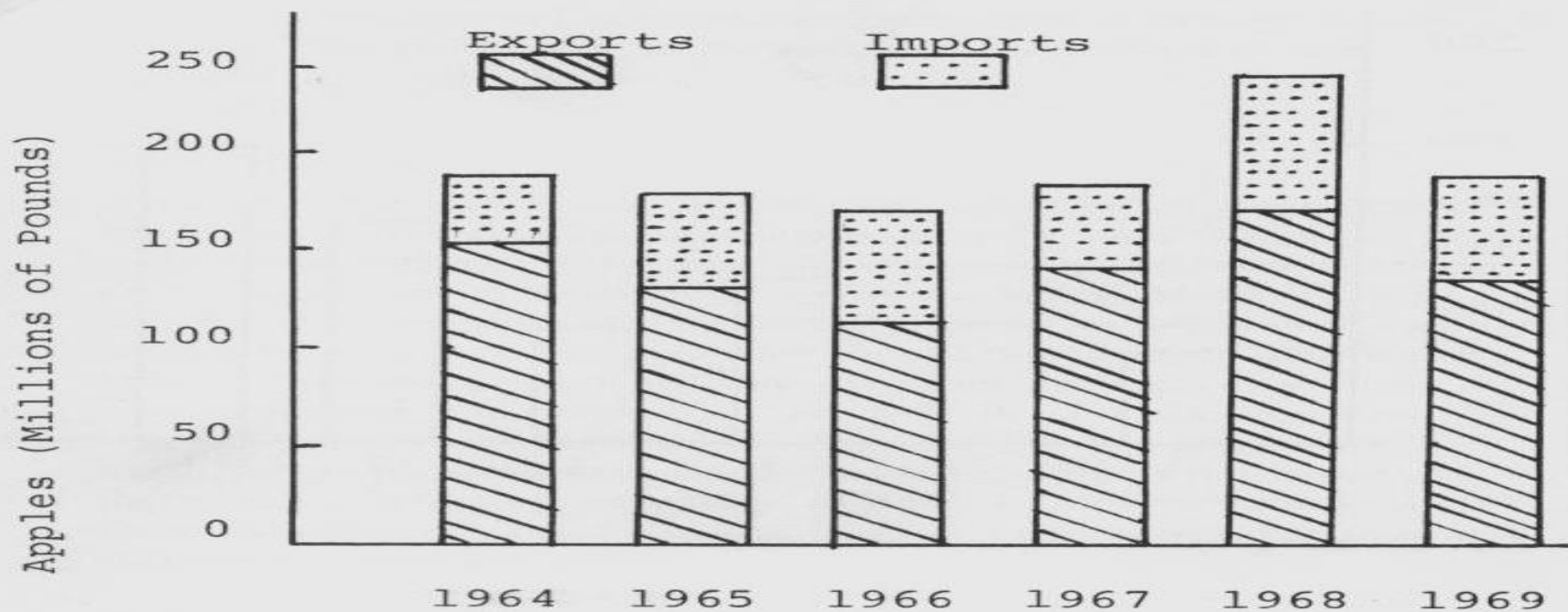


Fig 1. Component bar graph for import/export

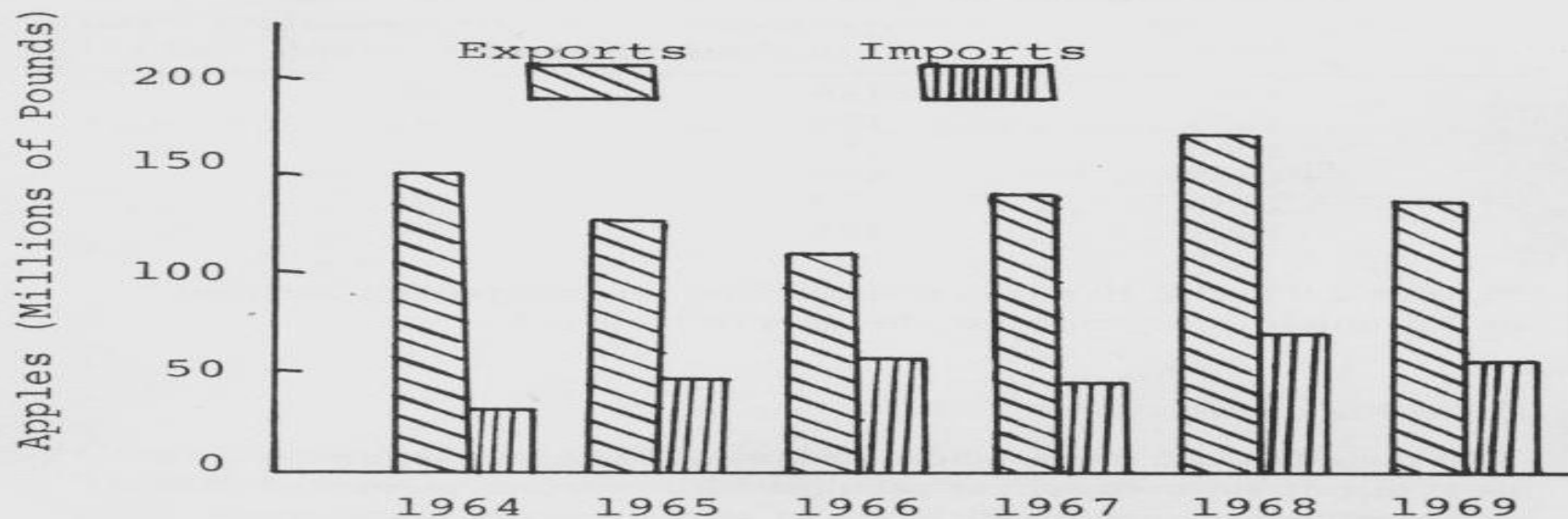
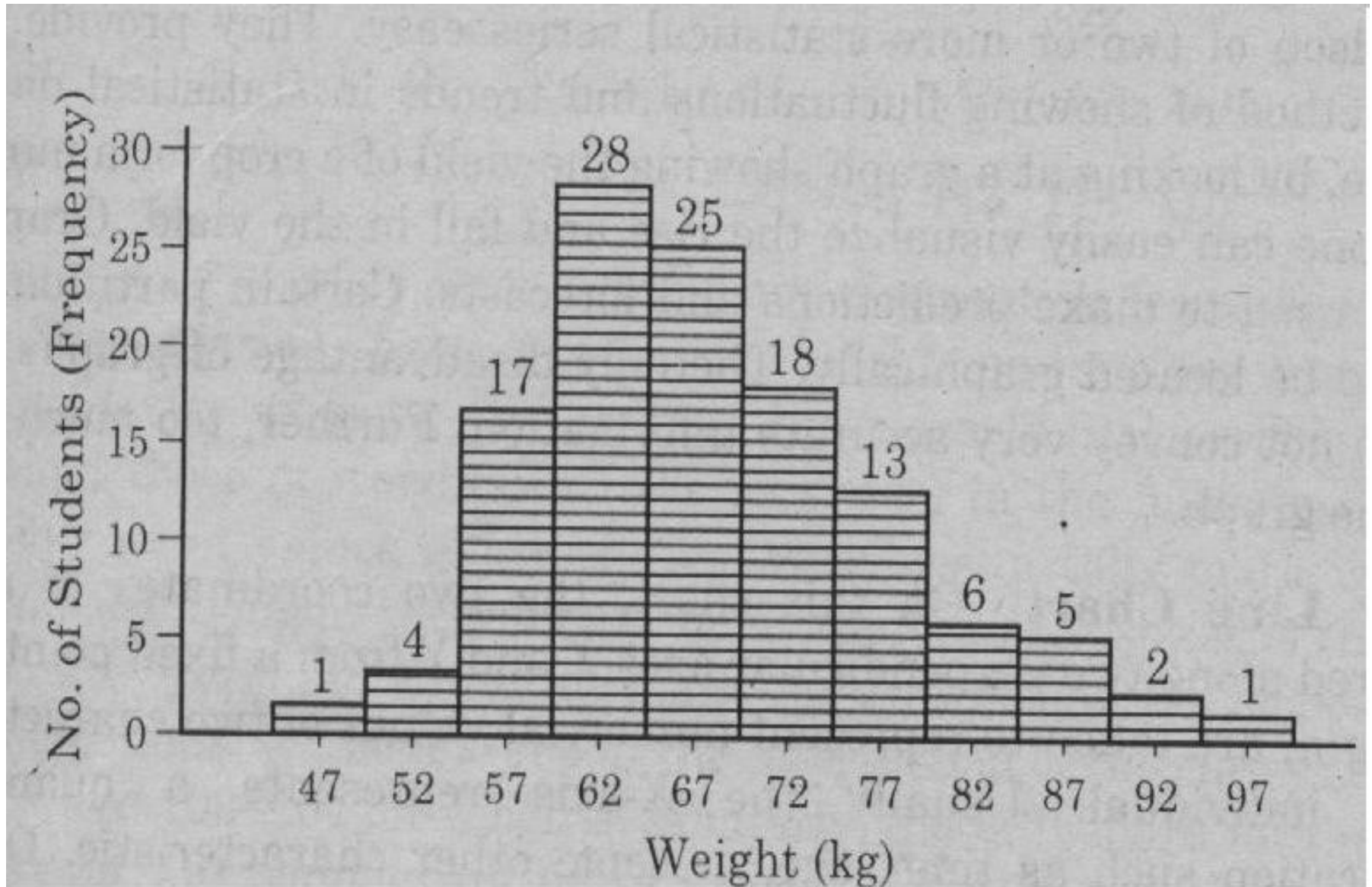


Fig 2: Multiple bar chart for export/import

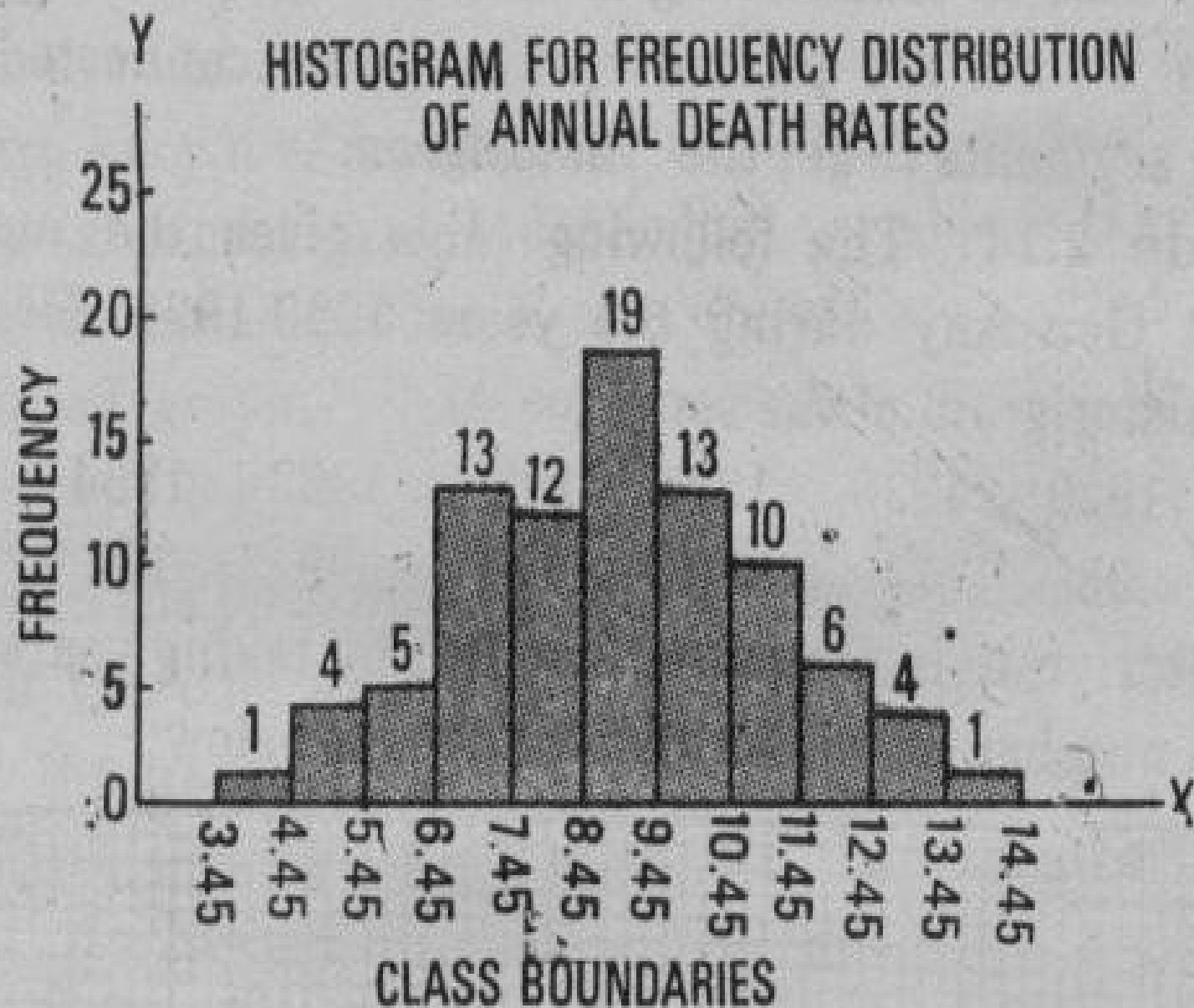
Histogram of weight of 120 students:



Definition of Statistics

- The word statistics seems to have been derived from the Latin word status or the Italian word statista. Both these words means a political state.
- **In plural sense**, statistics refer to numerical facts in any field of study, These facts are collected in systematic manner with a definite purpose in view. For instance, statistics of births & deaths, price statistics, agricultural statistics, statistics of crimes, statistics of road accidents, statistics of educational institutions etc.
- **In singular sense**. It refers to the science or discipline that includes procedures & techniques of collection, presentation, analysis & interpretation of numerical data.

HISTOGRAM FOR FREQUENCY DISTRIBUTION
OF ANNUAL DEATH RATES



Quadratic Equation

6.2.1 (a) Observed & Fitted proportion, $P(t)/n$, over the days for January

